

Advanced Natural Language Processing

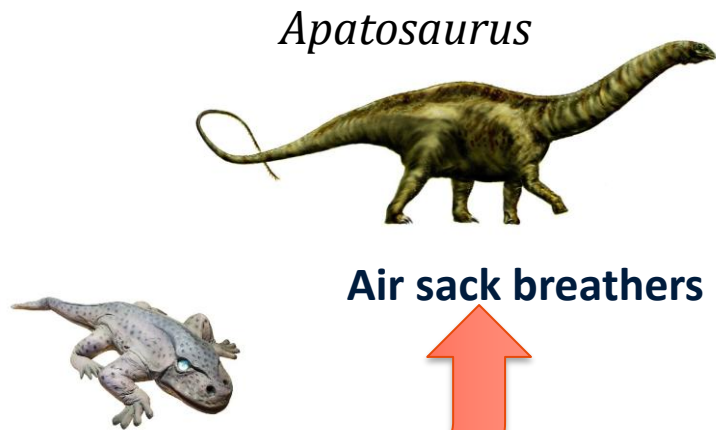
Spoken Language Processing

Prof dr ir Jan Scholtes and Aki Härmä
2026-2027

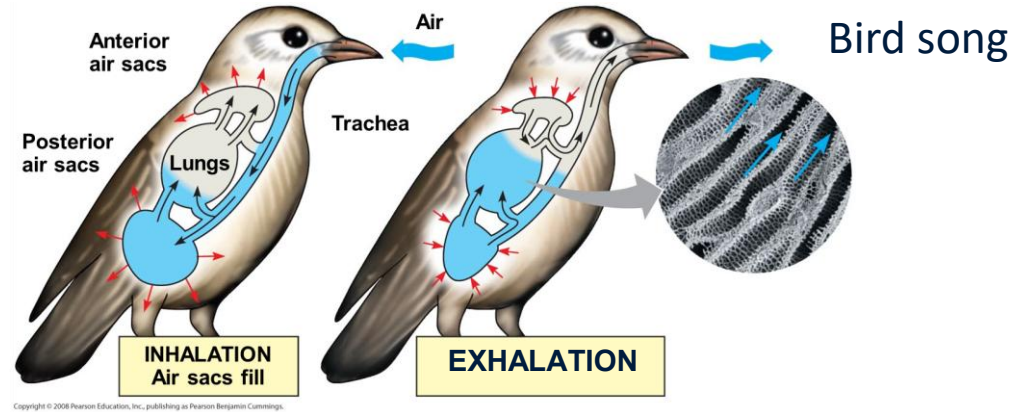


Maastricht University

Vocal communication – evolutionary foundations



Air sack breathers



530 M

350 M

150 M

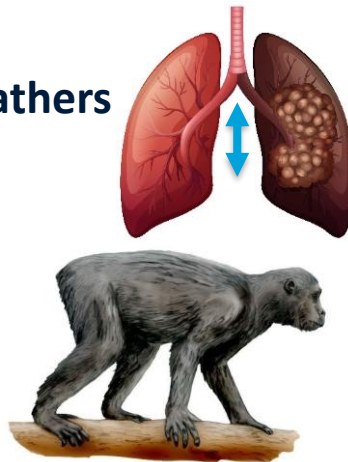
1.5 M

5000

Tidal breathers



Procynosuchus



Mammals,
vocal communication

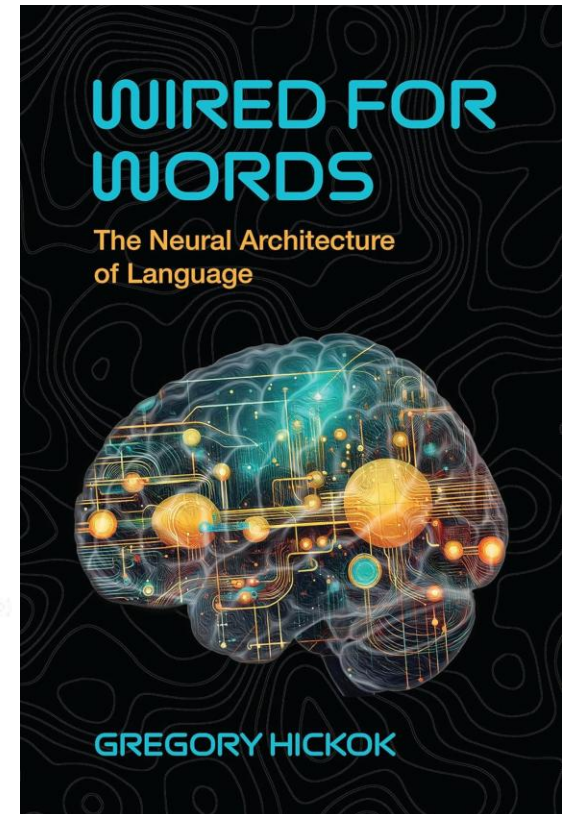
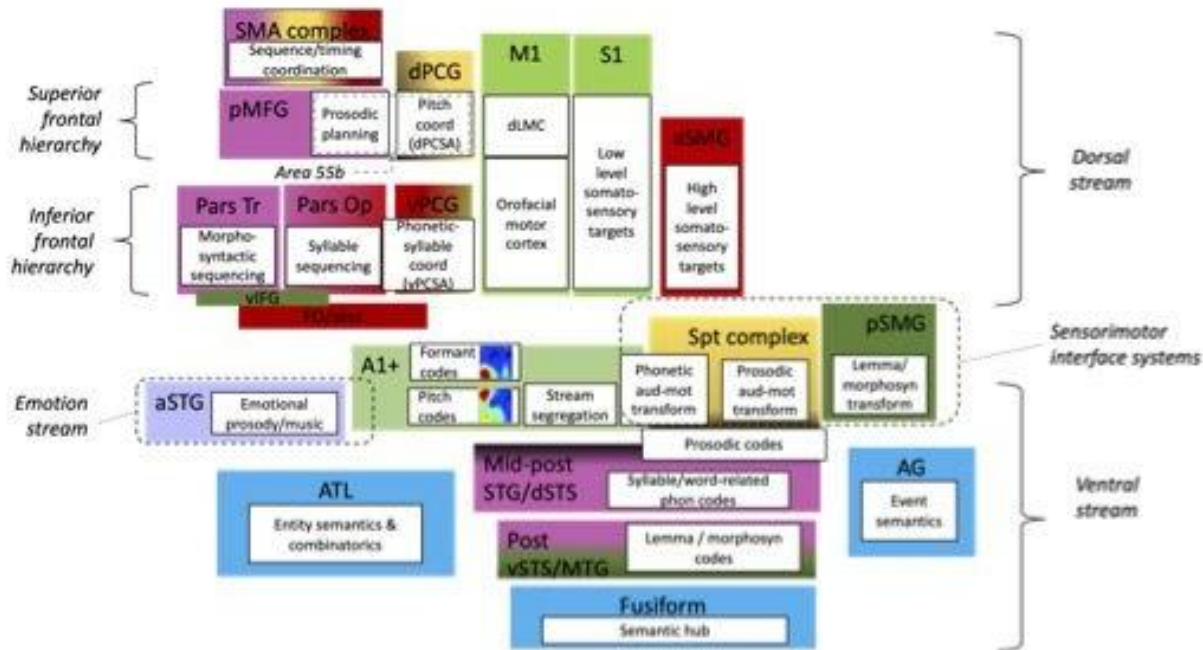


Homo erectus (1.5M) – spoken language

- Written language (3000 BC)
- Printing (1450 AD)
- Speech transmission (1876)
- Computer speech (1950)
- Large Language Models (2018)

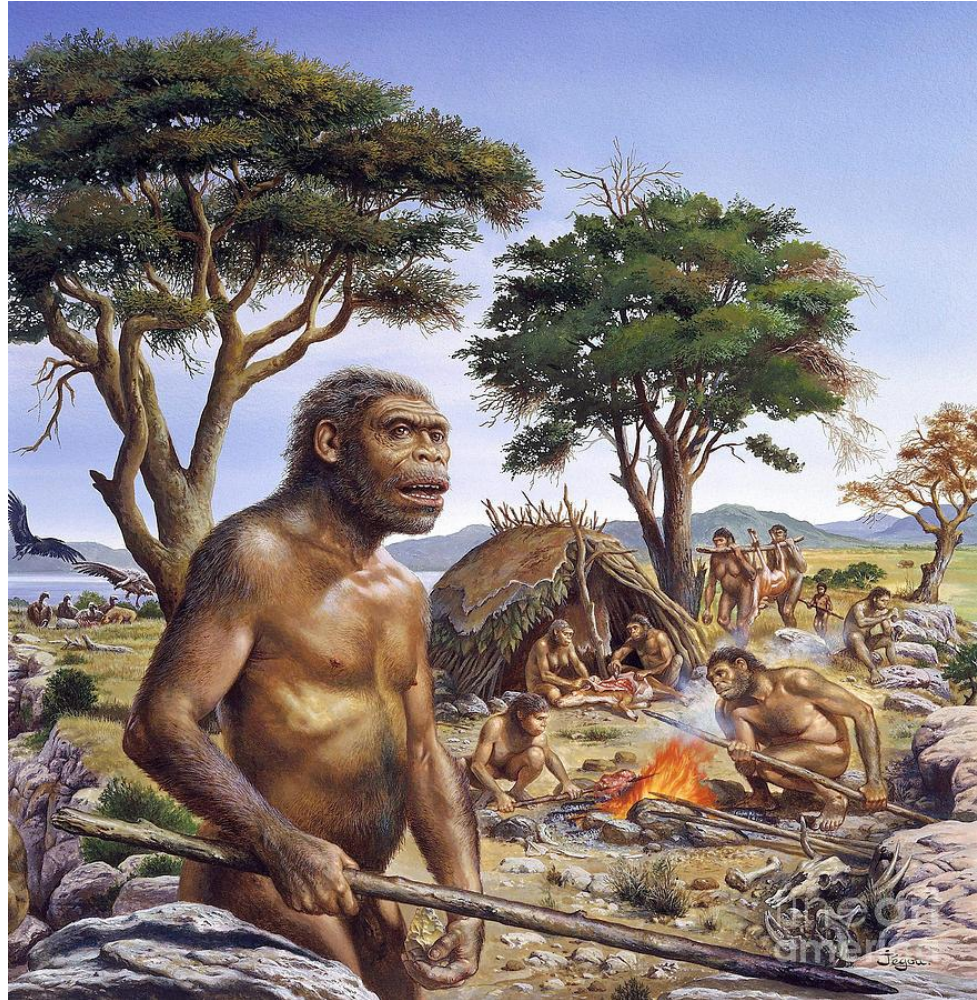
Linguistic-Sensorimotor Model (LSM)

Language system is a variant of the sensorimotor processing, like our movement



Speech – evolutionary foundations

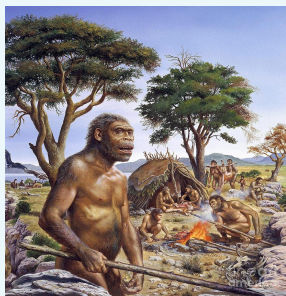
- Vocalization: adaptation of breathing for communication
 - First land animals (0.5G)
 - Icons and indices
- Spoken language
 - *Homo Erectus* (1.5M)*
 - symbols, complex behavior, learning, culture
- Further evolution of speech organs in homini:
 - *Homo Sapiens* (0.3M)



Natural Languages and Language Processing

SPEECH, SPOKEN LANGUAGE

- 1.5 million years ago
- Speech recognition, transmission, analysis, synthesis, end-to-end speech translation



(HAND-)WRITING, PRINTED TEXT

- Sumer 3500 BC
- Printing in 6th century China
- Transcription, Optical Character Recognition (OCR), hand-writing analysis.



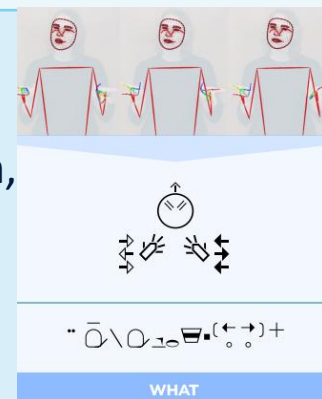
TEXT PROCESSING (this course)

- Linguistics: India 600 BCE, last 150 years
- Parsing, classification, information retrieval, text analysis, NLP/G/U



SIGN LANGUAGE

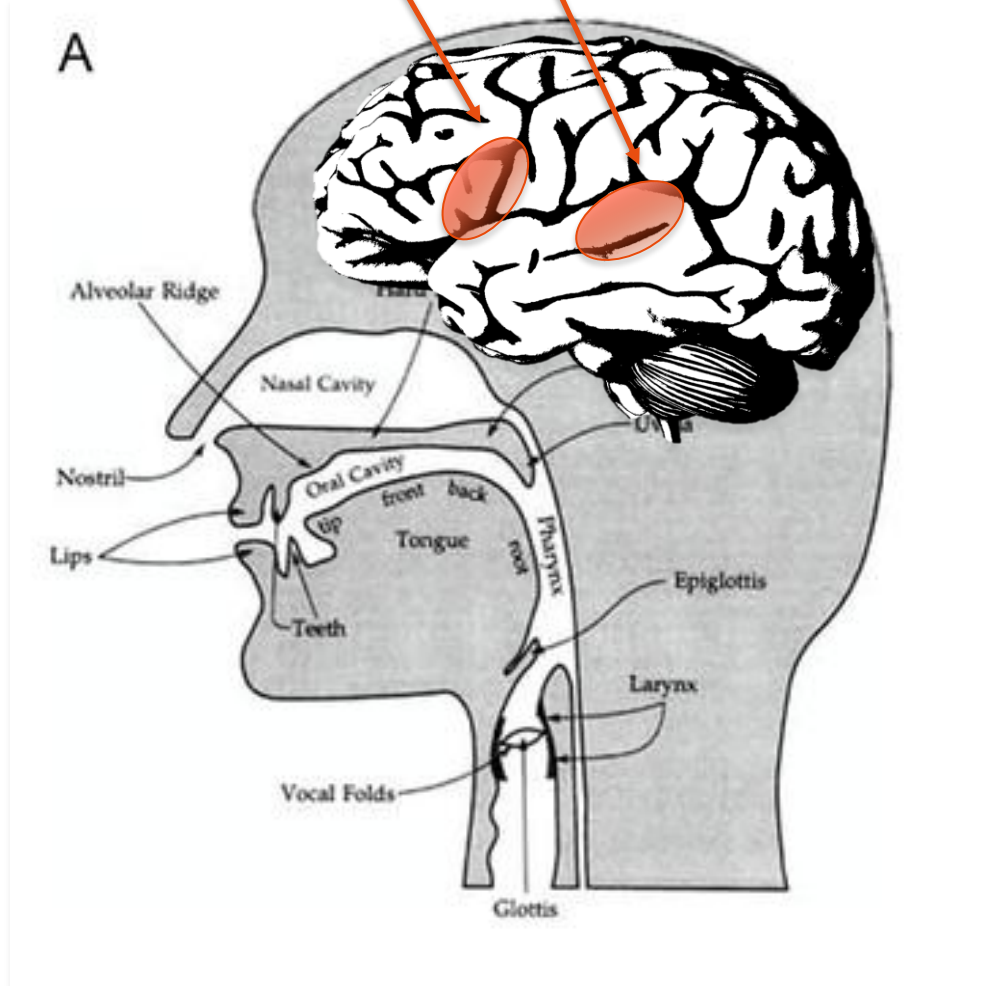
- Greece 400 BCE (known)
- Transcription, translation, synthesis (animation)
- A fairly new field (after 1980's, 2000's)



Speech production

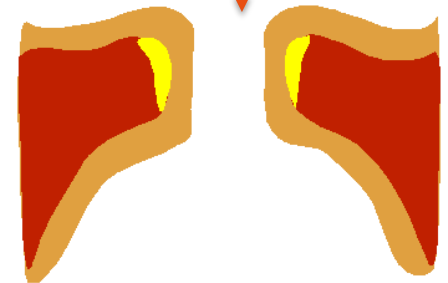
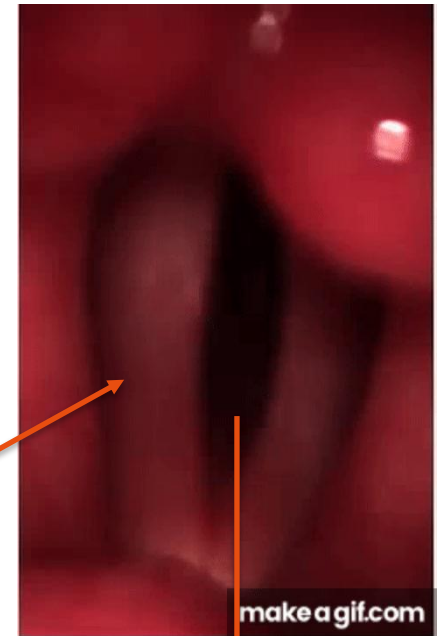
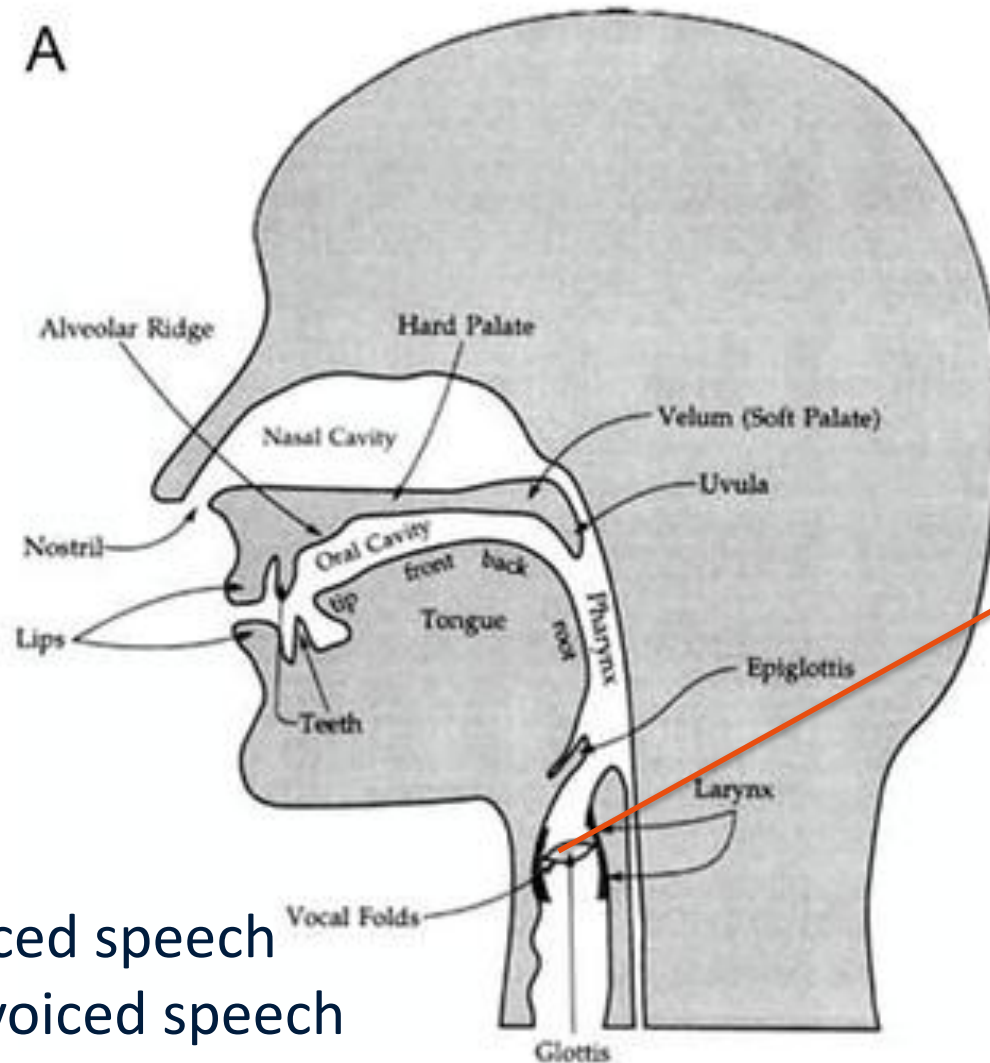
- Speech is
 - breathing modulated by linguistic content
 - neural control of muscles of the vocal tract
 - control of a glottal excitation and the shape of a resonant tube

Broca's area – speech production
Wernicke's area
– speech perception



Speech production: glottal excitation in vowels

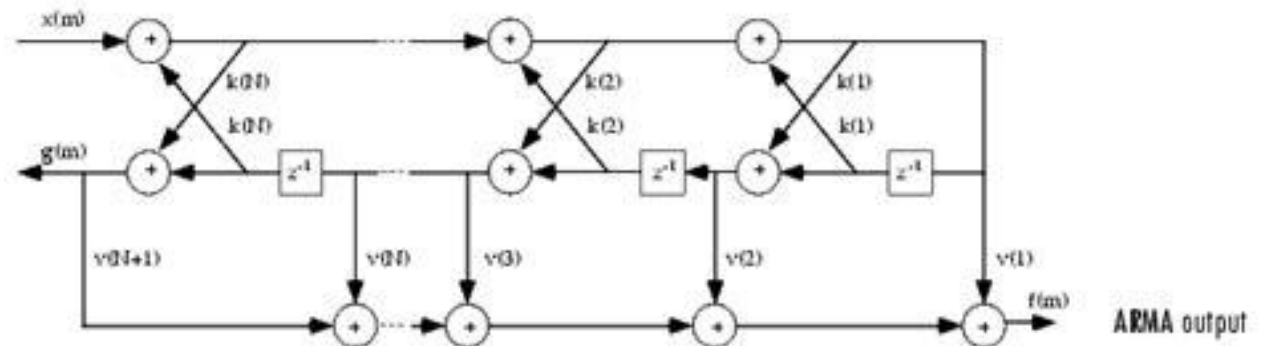
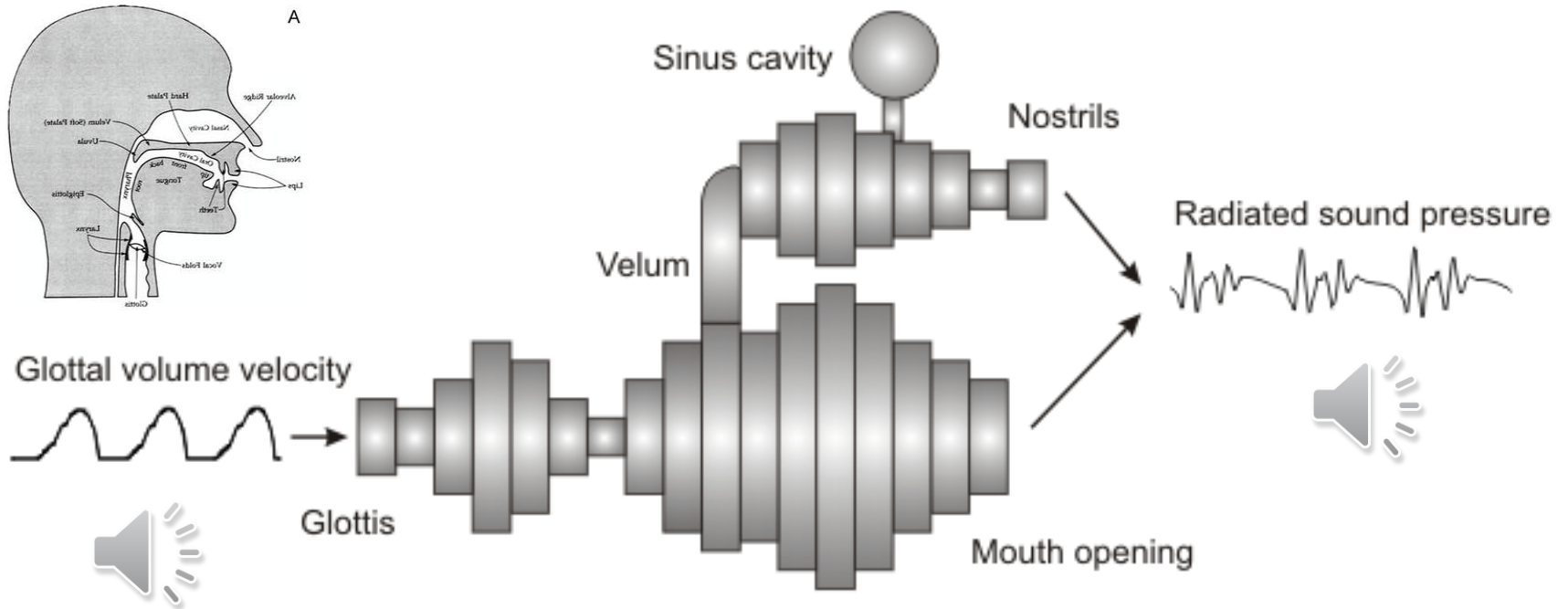
A



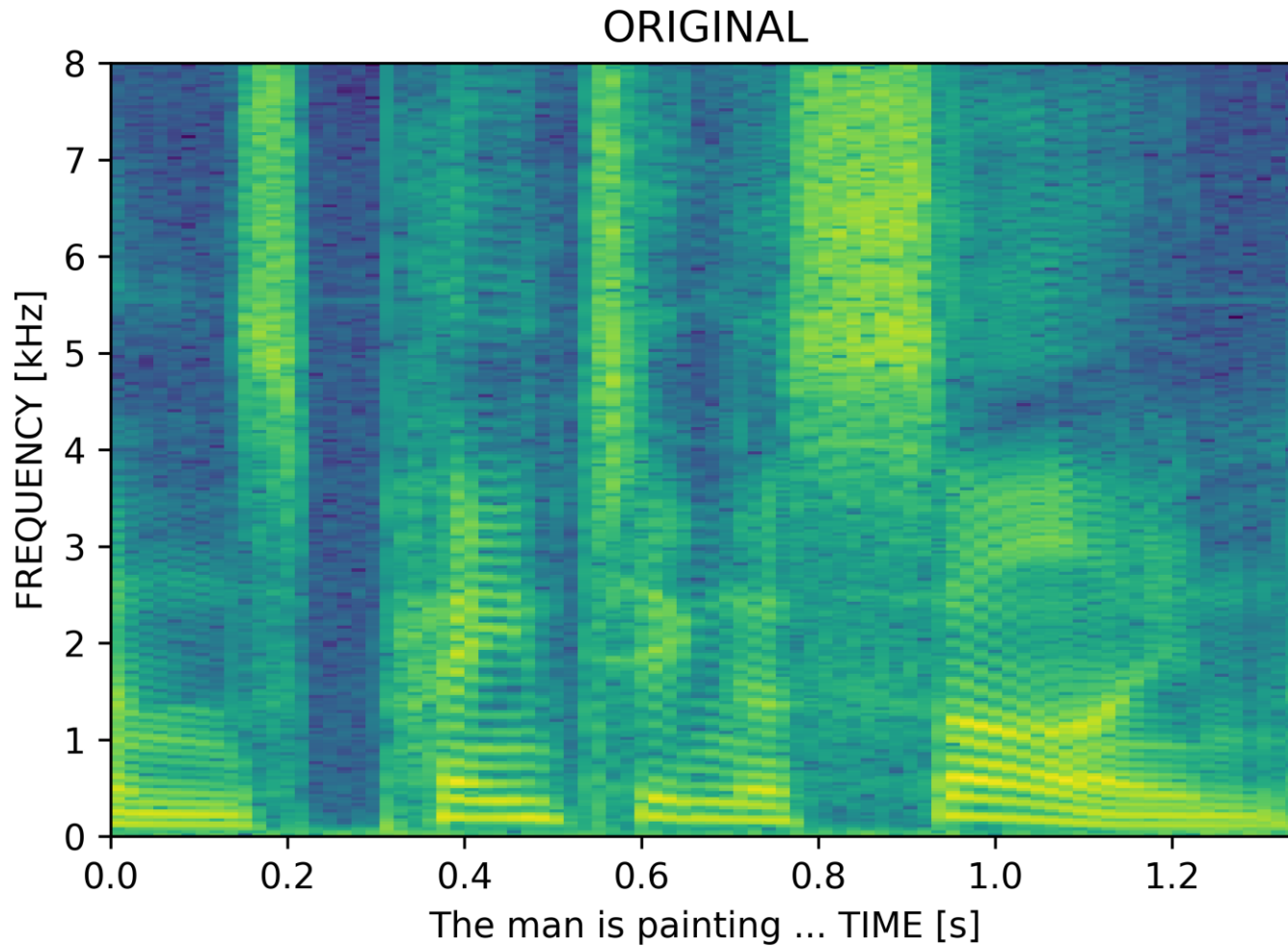
- Voiced speech
- Unvoiced speech

F0 - 40-400 Hz

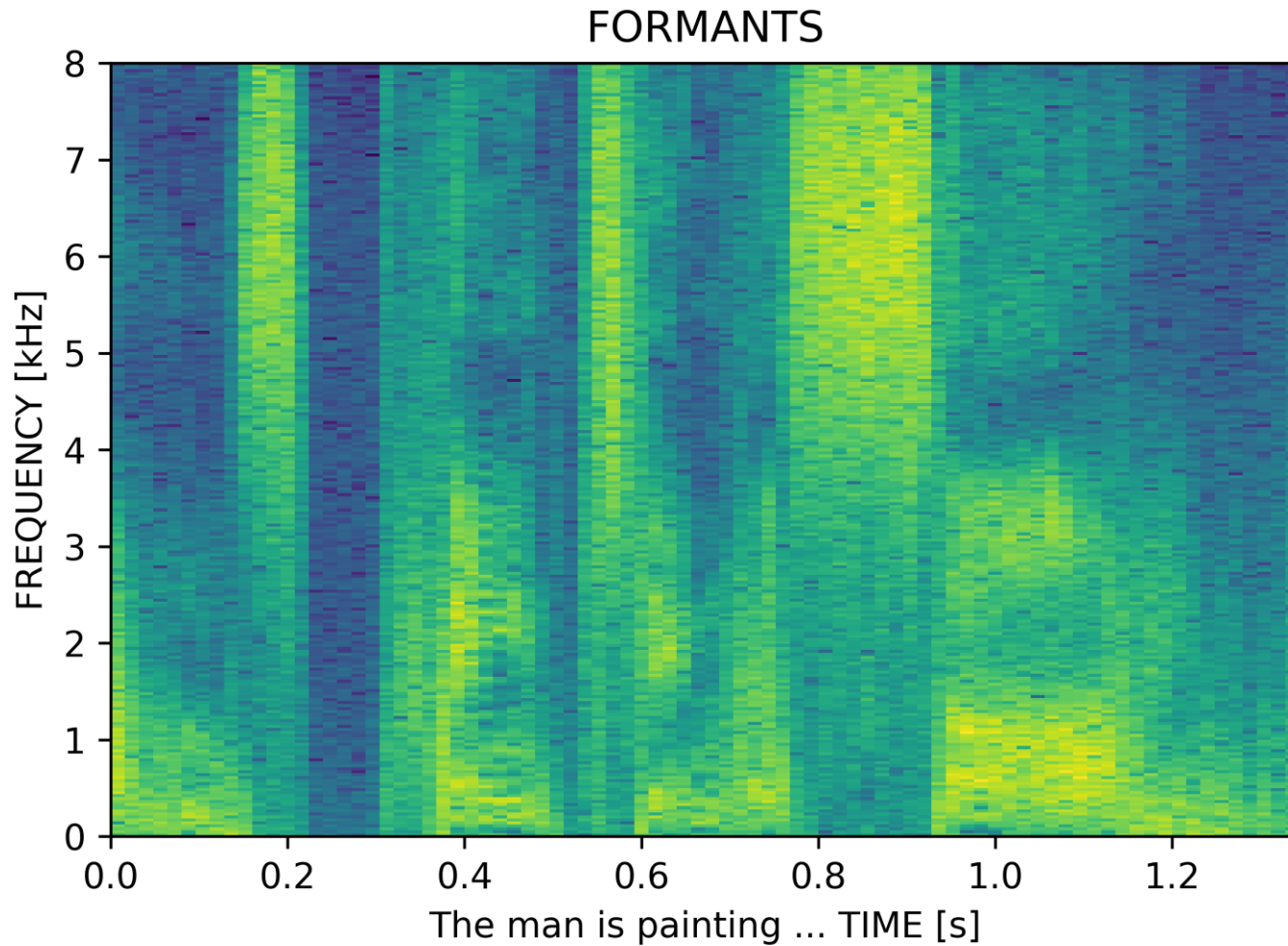
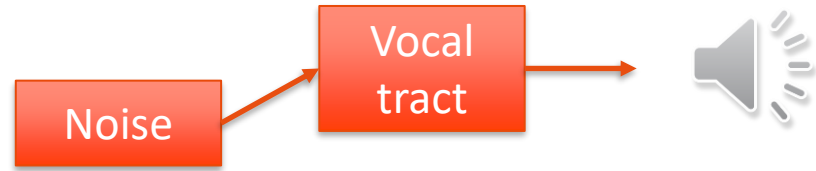
Speech production: vocal tract model



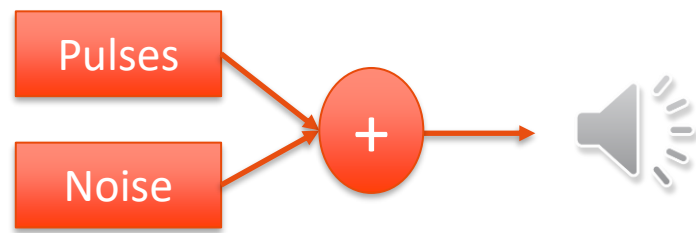
Speech spectrum



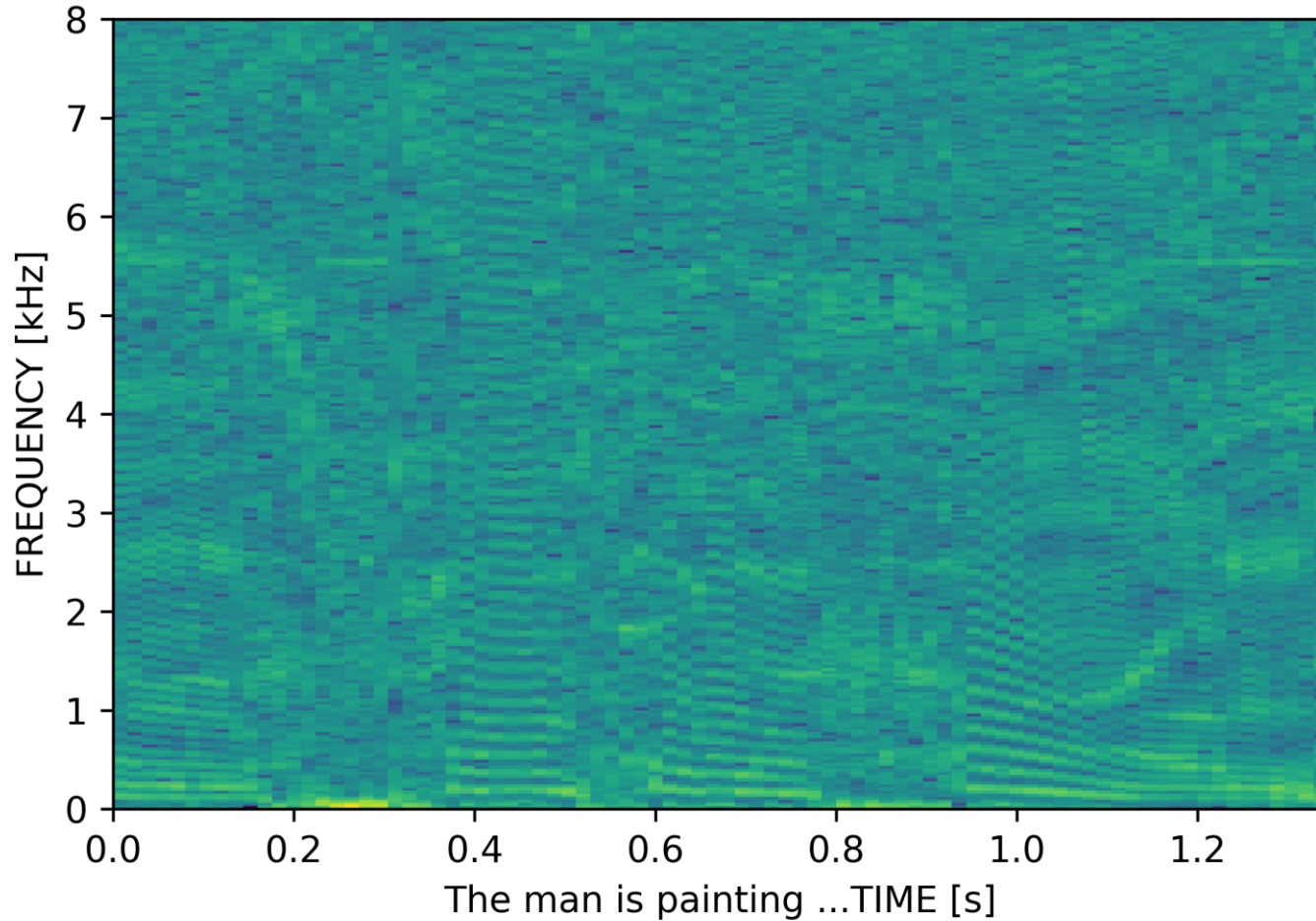
Speech spectrum



Speech spectrum



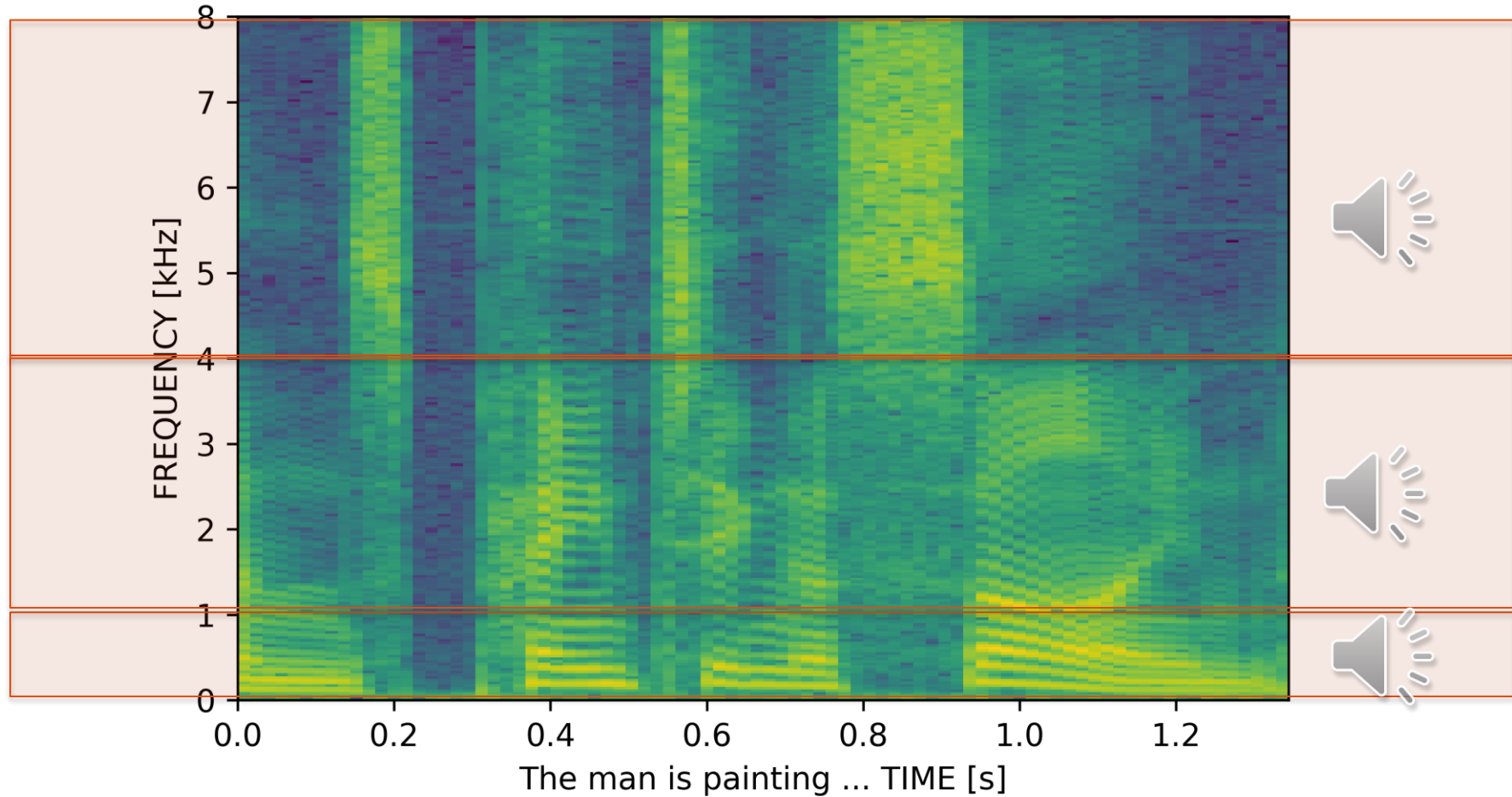
EXCITATION



Speech spectrum



ORIGINAL

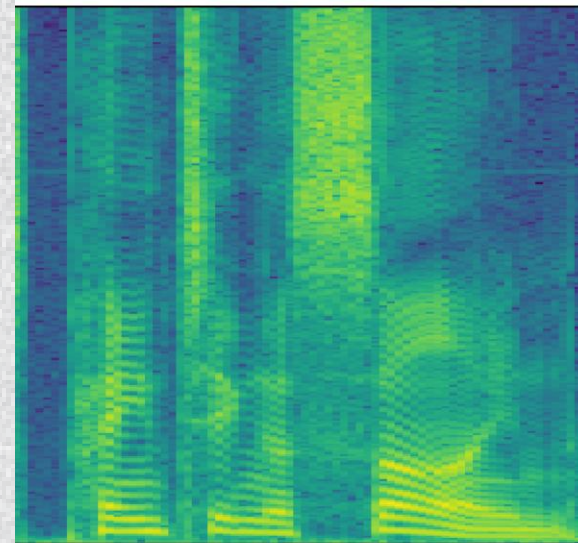


Nothing gold can stay.

Robert Frost (1923)

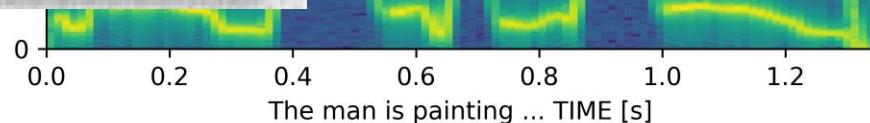
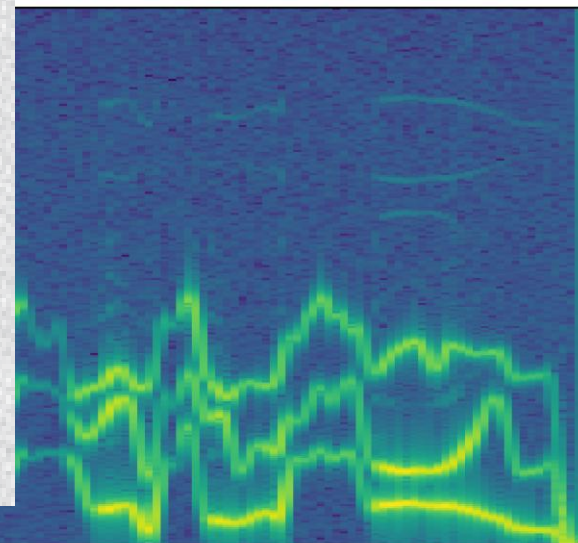
Nature's first green is gold,
Her hardest hue to hold.
Her early leaf's a flower;
But only so an hour.
Then leaf subsides to leaf.
So Eden sank to grief,
So dawn goes down today.
Nothing gold can stay.

Full spectrum speech



... TIME [s]

usoid-in-formants speech



The man is painting ... TIME [s]

Speech representation

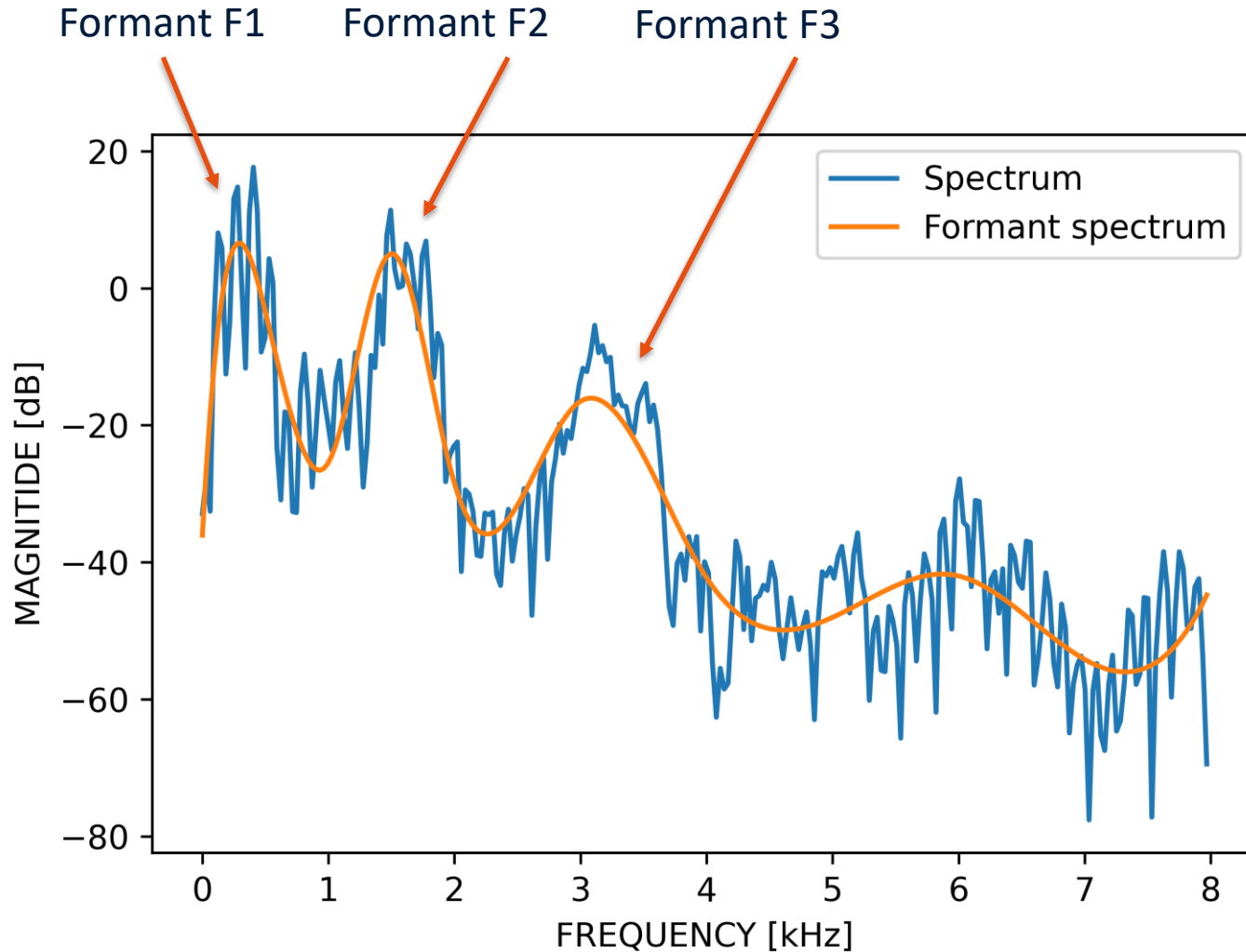


Maastricht University

Spectrum formants

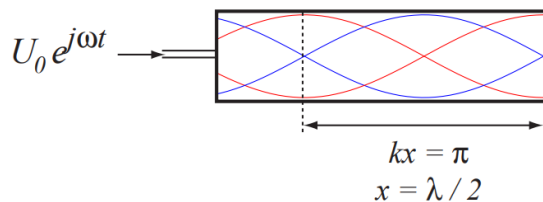
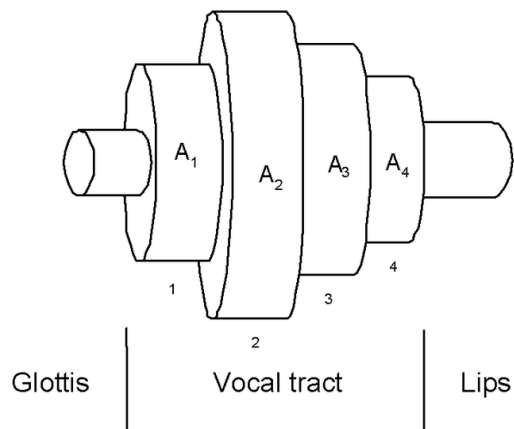
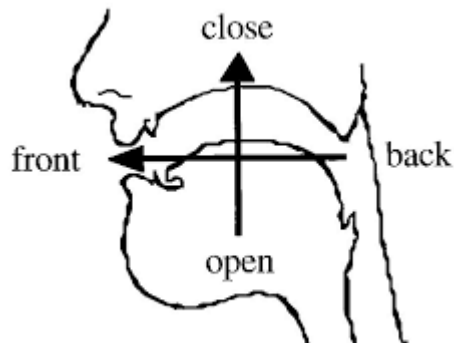
Formants – vocal tract

Fine-structure - the excitation



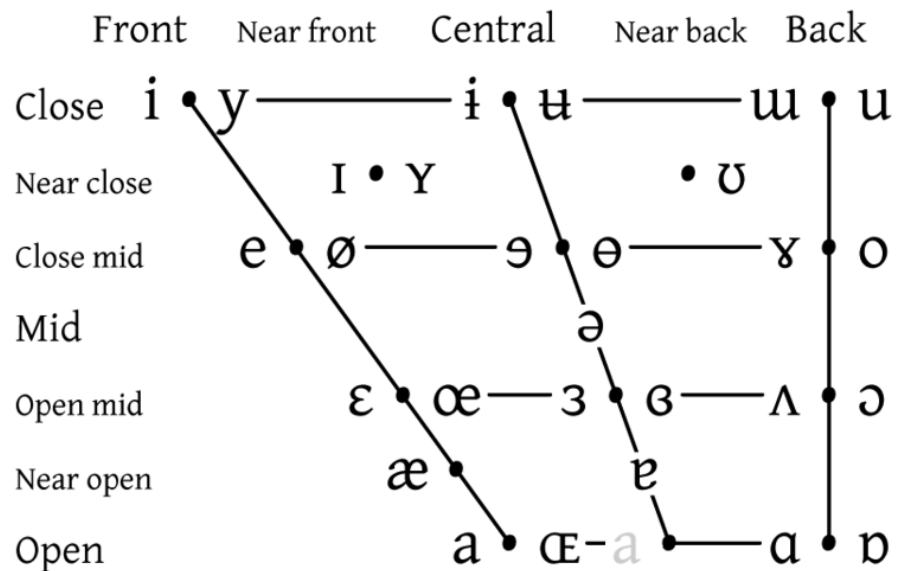
Vowel Space – voiced speech

Vocal tract



pressure = 0 (node)
vol. veloc. = max (antinode)

VOWELS

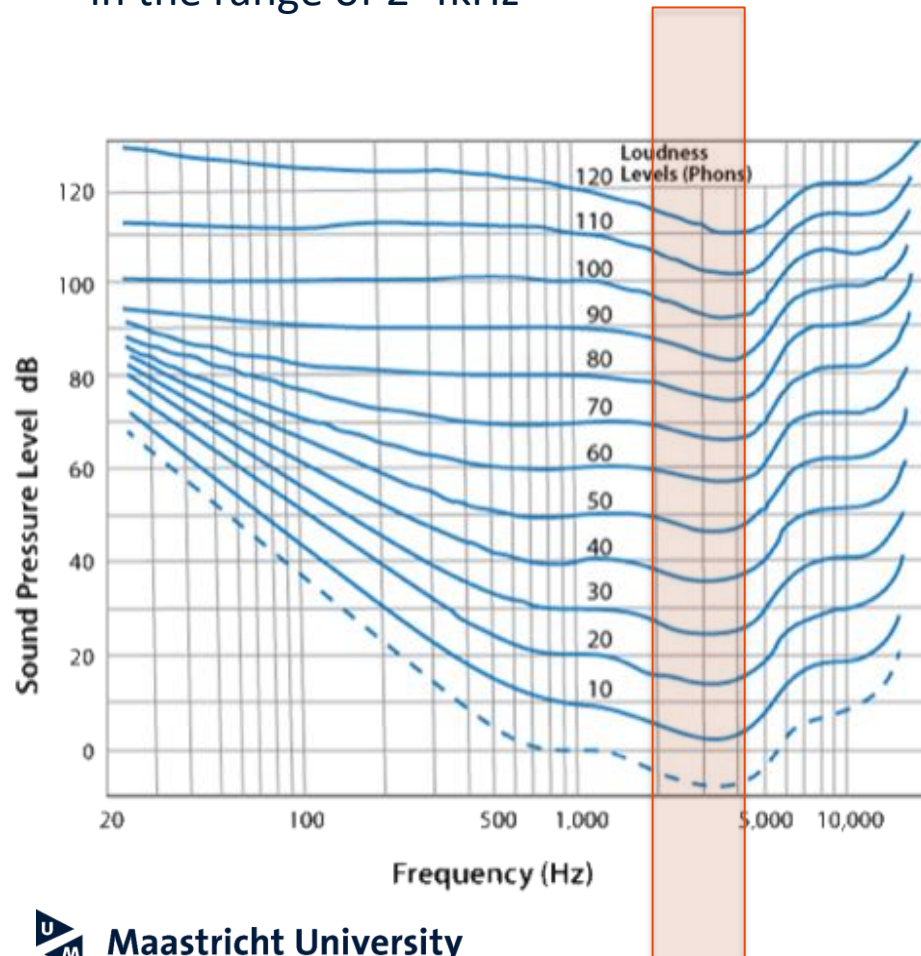


Vowels at right & left of bullets are rounded & unrounded.

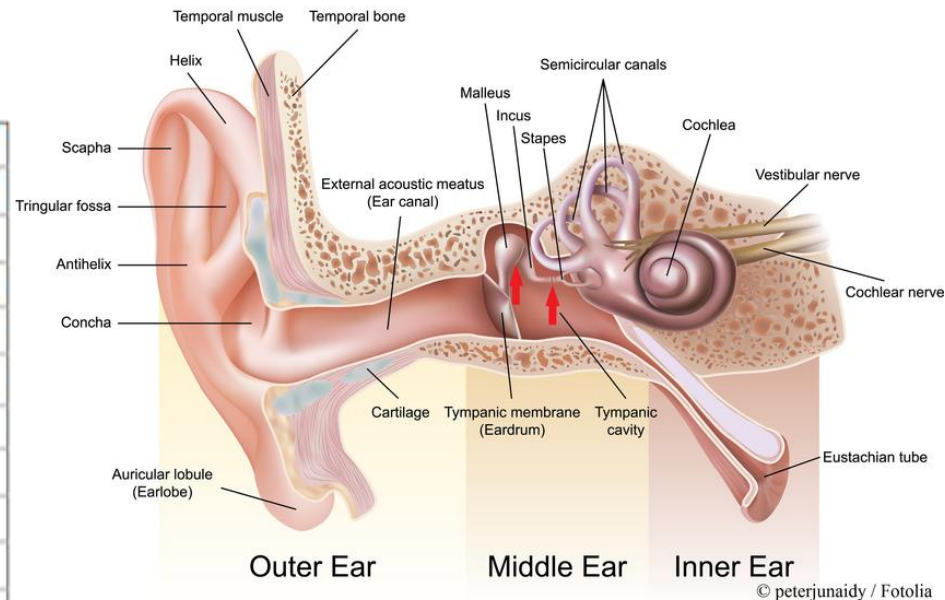
IPA – International Phonetic Alphabet

Hearing – the receiver system for speech

The ear canal resonance amplifies sounds in the range of 2-4kHz



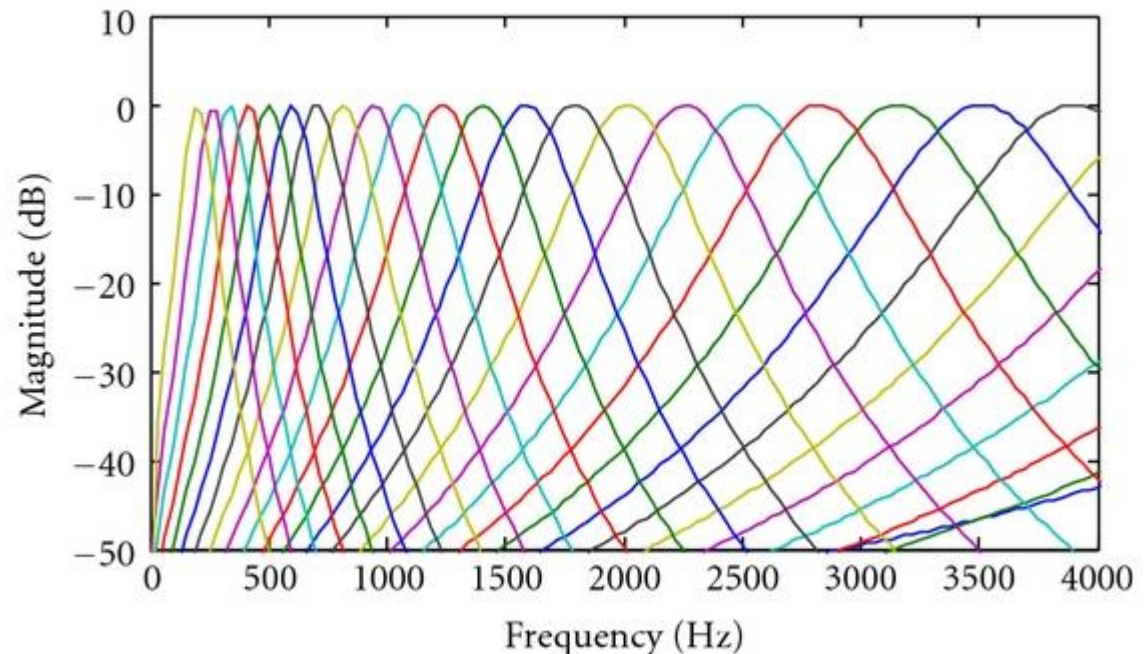
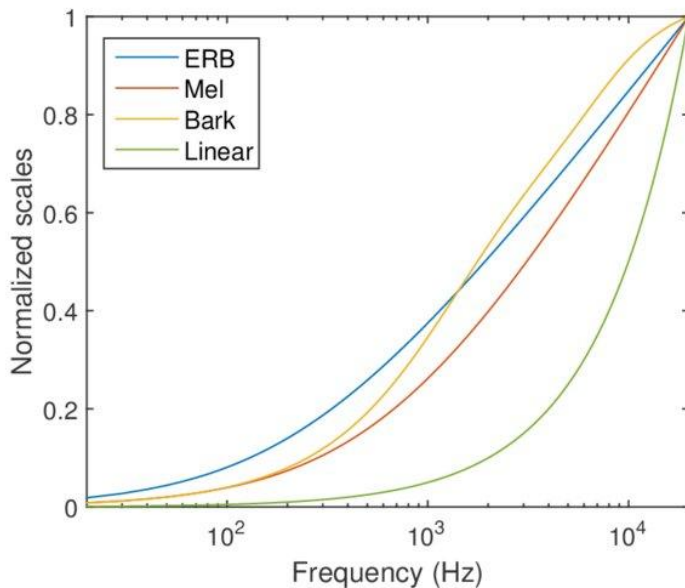
Anatomy of the Ear



Inner ear – Cochlea-to-neural transduction: Perception of loudness is logarithmic (decibel scale)

Auditory filterbanks

- Mel - scale: based on perceived pitch differences (Stevens and Volkmann, 1940)
 - Widely used in speech processing due to historical reasons
- Bark - scale: based on “Critical Bands” of hearing (Zwicker, 1961)
- ERB - Equivalent Rectangular Bandwidth (Moore & Glasberg, 1984)
 - More plausible scale based on more recent research (but more log-like)
- Gammatone filters (roughly ERB filterbank with the correct temporal response)

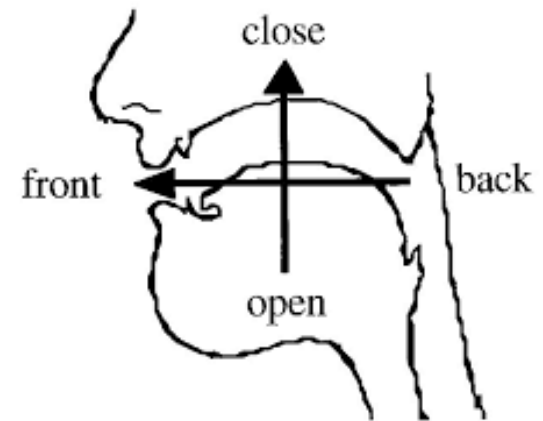


Human Speech: Phonemes/Phones

- **Phones:**
 - Basic speech sounds (around 200-300)
- **Phonemes:**
 - Defined as fundamental units of speech
 - If you change it, it (can) change the meaning
 - Language dependent
- **Vowels – voiced speech**
 - Voiced speech, glottis exciting open vocal tract
 - There are six vowels in the English language: a, e, i, o, u and sometimes y.
- **Consonants – unvoiced speech:**
 - sound segments in which the airflow is at least partly obstructed
 - The 21 consonant letters in the English alphabet are B, C, D, F, G, H, J, K, L, M, N, P, Q, R, S, T, V, X, Z, W and Y.

Coarticulation and assimilation

- Speech sounds change in the linguistic context
 - Individual syllables in speech <200ms
 - Movements of our tongue and jaw >200ms
- Coarticulation
 - /k/ in /kier/ or /koel/
- Assimilation
 - /d/ in /good boy/ or /good girl/



Prosody and paralinguistics

Pulses

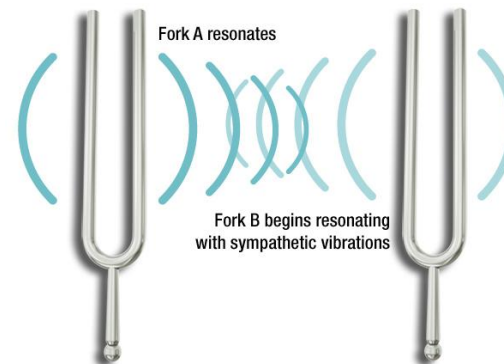
Noise

- Pitch
- Loudness
- Rhythm
- Intonation

Word	Symbols	Tone names	Meaning
Yes	→	Level	Neutral; uninterested: 'Yes'
Yes	↘	Fall	Positive response 'I agree'
Yes	↗	Low-rise	Question, eliciting a response
Yes	↑	High-rise	Disappointment: 'Are you saying Yes'
Yes	↘↗	Fall-rise	'Carry on, I'm listening'
Yes	↗↘	Rise-fall	Reserved, indicating doubt

In Conversation

- Back-channeling
- Entrainment
- Barge in

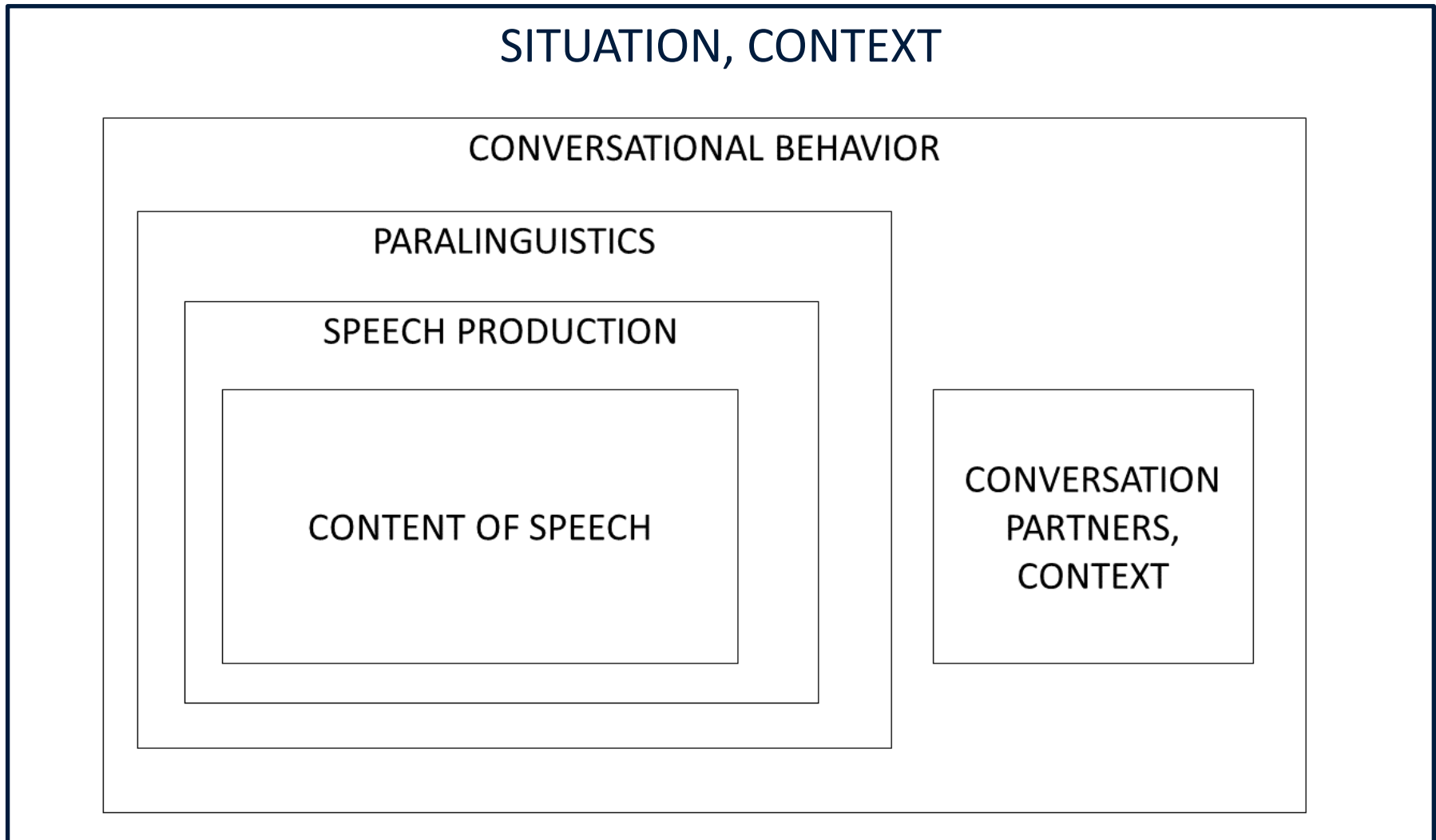


Speech vs text

Channel	Words per minute (English)
Speaking (vocal track muscles)	Which is fastest and which slowest?
Listening (hearing neural proc.)	
Typing (hand & finger muscles)	
Reading (vision neural proc.)	

- The fastest information channel for human-to-human communication is **speaking** combined with **reading**
- But, the channel from speech-to-text is a very lossy: Prosody, emotions, personality,... tons of bits/s .. Lost!

What is in the speech?



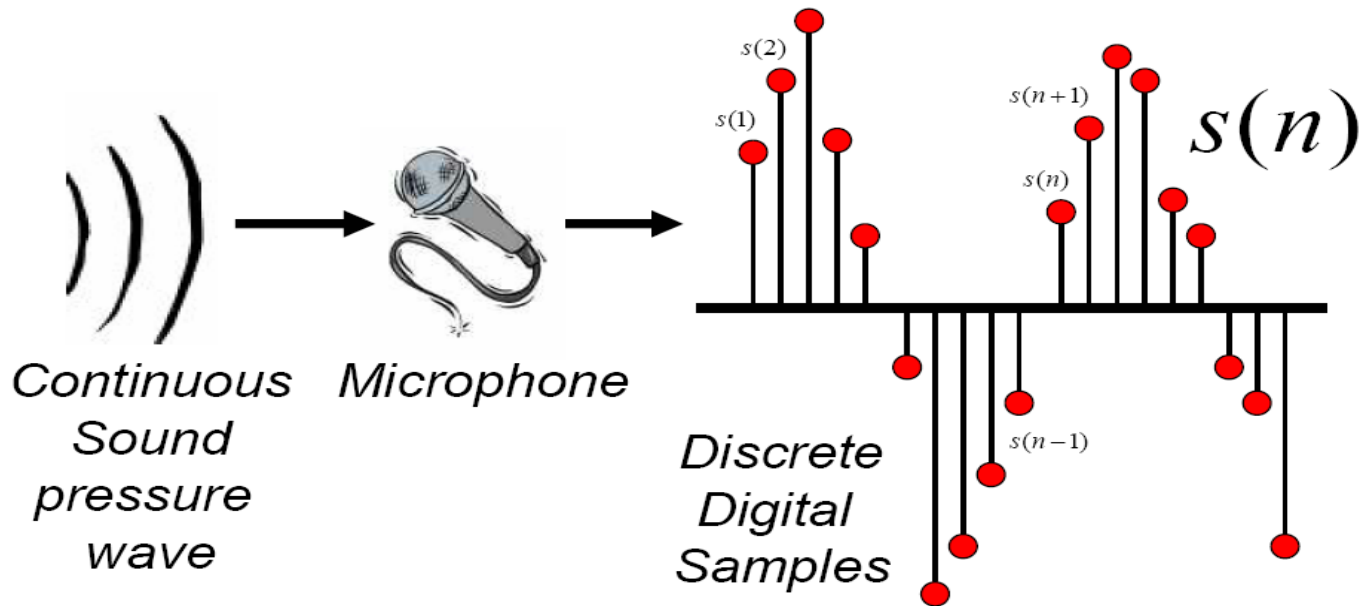
Spoken language applications

- Speech capture and enhancement
- Speech transmission and coding
- **Automatic Speech recognition (ASR)**
- Speaker identification / verification
- Speaker state/gender/health/... detection
- Speech language/dialect detection
- Speech synthesis
- **Speech-to-Speech, Speech-to-X**

We could have a separate module on each of these topics

Audio signal

- **Sampling:**
 - measuring amplitude of signal at time t
 - 44.1 kHz or 48kHz Audio books
 - 16 kHz (samples/sec) wideband speech (zoom/Teams calls etc):
 - 8 kHz (samples/sec) Conventional telephone speech



Classic ASR systems (1980's-to-2010's)



Fast Fourier Transform (FFT) - speech to spectrum

Mel-spectrum mapping (and Mel-cepstrum by another FFT) - to auditory representation, Decibel magnitude (logarithm), or cepstrum

TRAINED WITH LARGE COLLECTION OF TRANSCRIBED SPEECH DATA

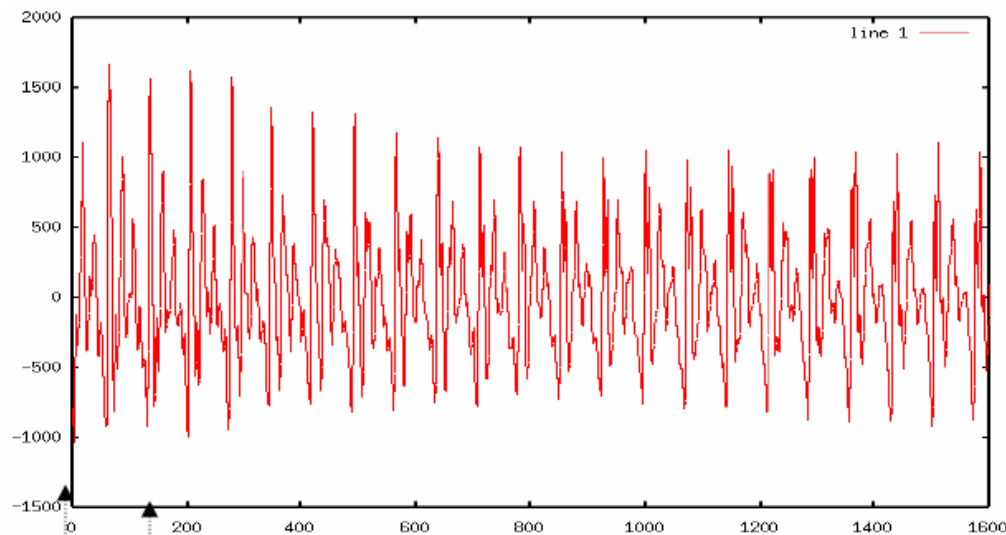
Model spectra by Gaussian Mixture Models (GMMs) – model spectrum “states”

Hidden Markov Model (HMM) – model state transitions by a probabilistic model

Decoding of the most likely phoneme sequence by Viterbi algorithm

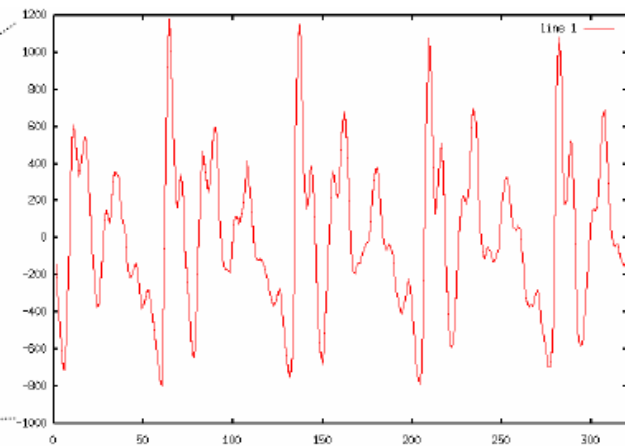
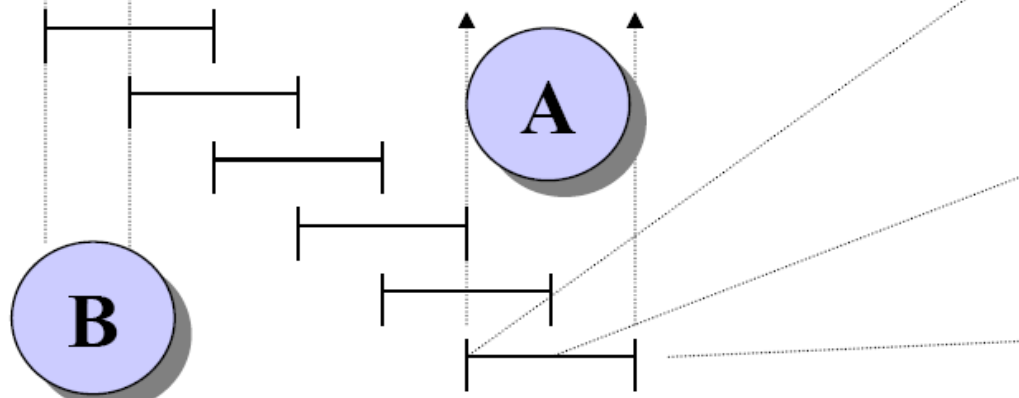
“The man is painting a sign”

Windowing



A $\sim 20 - 25$ ms

B ~ 10 ms

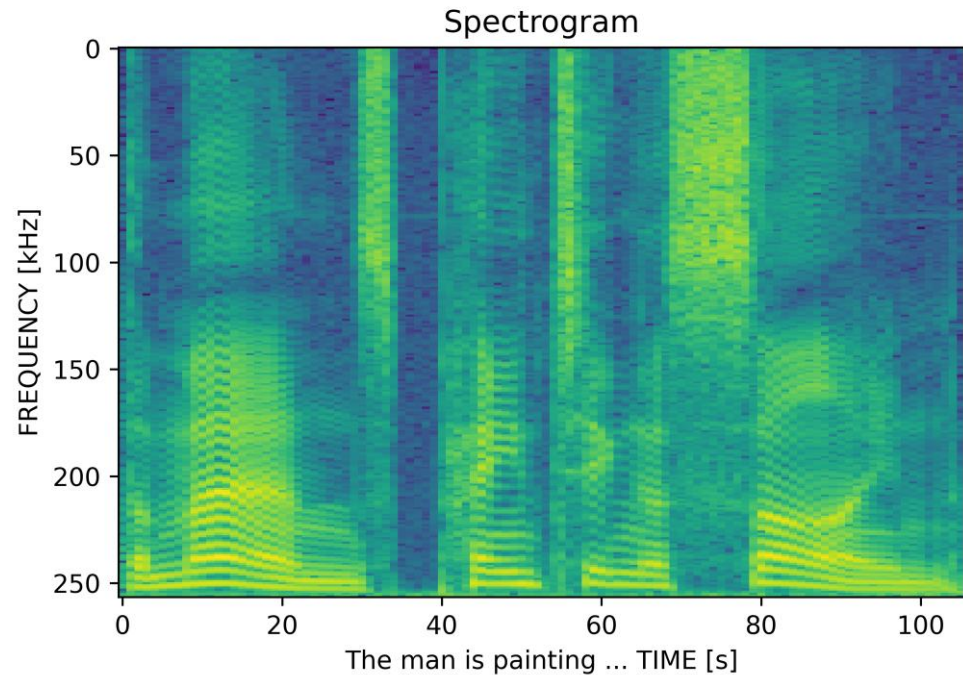


Why divide speech signal into successive overlapping frames?
Speech is not a stationary signal; we want information about a small enough region that the spectral information is a useful cue.

A Spectrogram

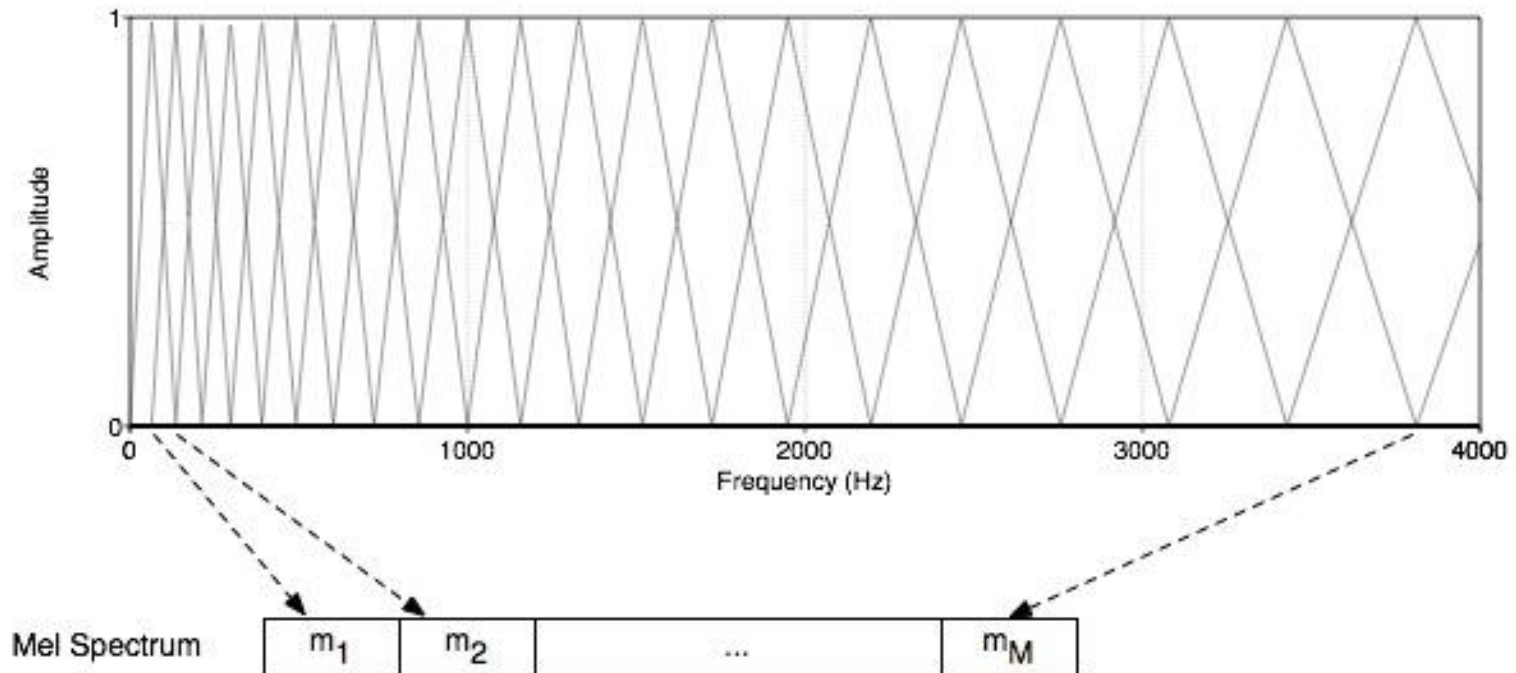
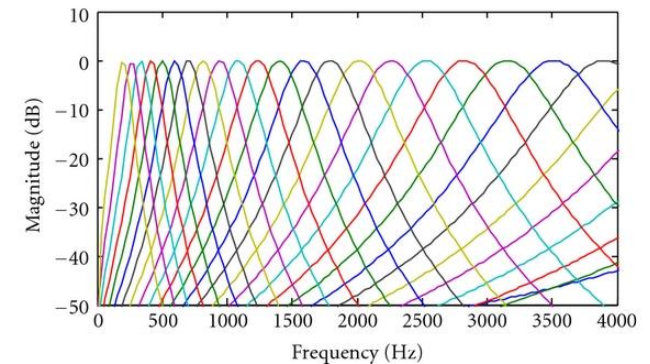
- A spectrogram represents a number of FFTs stacked on top of each other.
- It is a way to visually represent a signal's loudness, or amplitude, as it varies over time at different frequencies.
- The y-axis is converted to a log scale, and the color dimension is converted to decibels (e.g. the log scale of the amplitude).

Intuition: humans can only perceive a very small and concentrated range of frequencies and amplitudes.



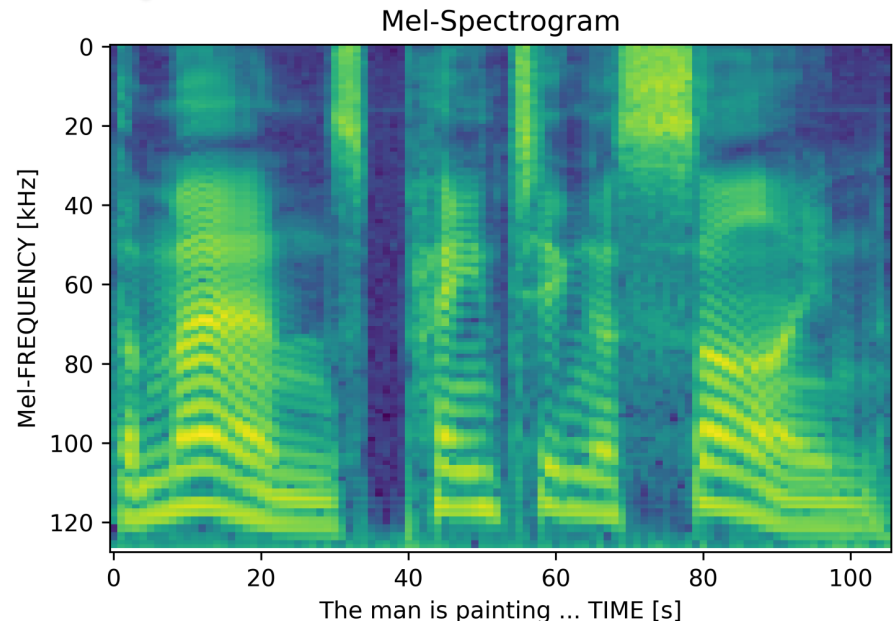
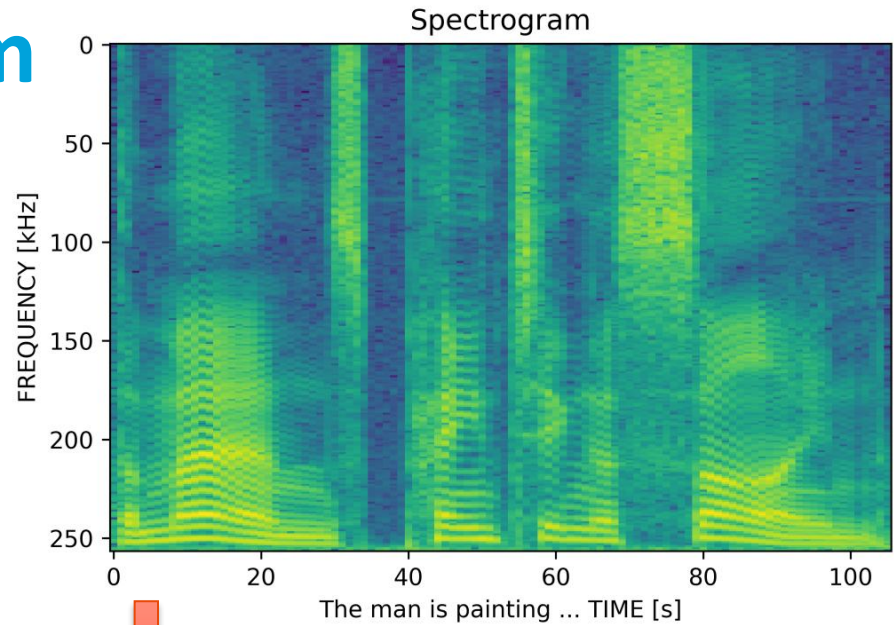
Mel Filter Bank Processing

- Mel Filter bank
 - Uniformly spaced before 1 kHz
 - logarithmic scale after 1 kHz



Log Mel Spectrogram

- A Mel spectrogram is a spectrogram where the frequencies are converted to the Mel scale.
- Spectrum magnitudes are mapped to a logarithmic Decibel scale
- Aka the **Log Mel Spectrogram**
- Matches with:
 - Frequency resolution and loudness perception of hearing
 - Harmonic spectrum of speech production



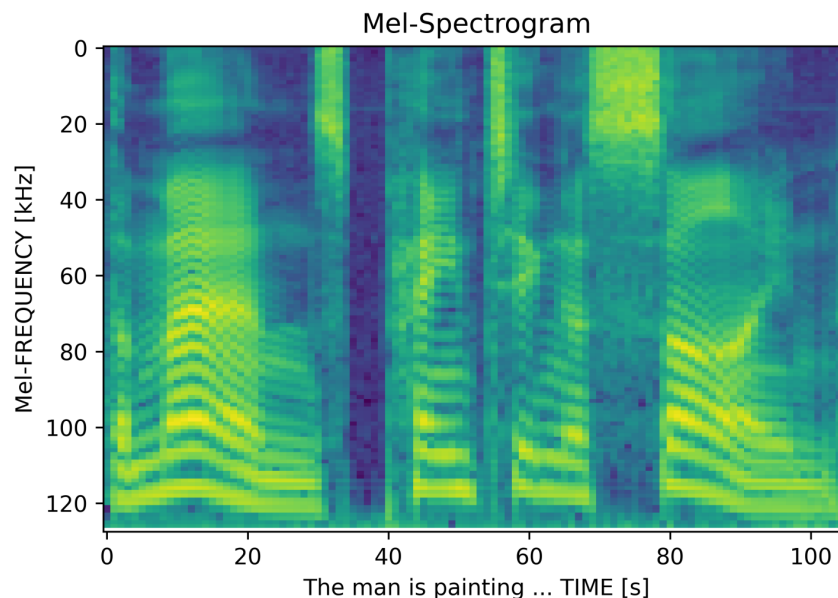
Data compression before NN



- HQ speech at 48kHz: 48000 numbers/s
- Down-sample to 16kHz: 16000 numbers/s
- Mel-spectrum (64 bands, 10ms steps): 6400 numbers/s

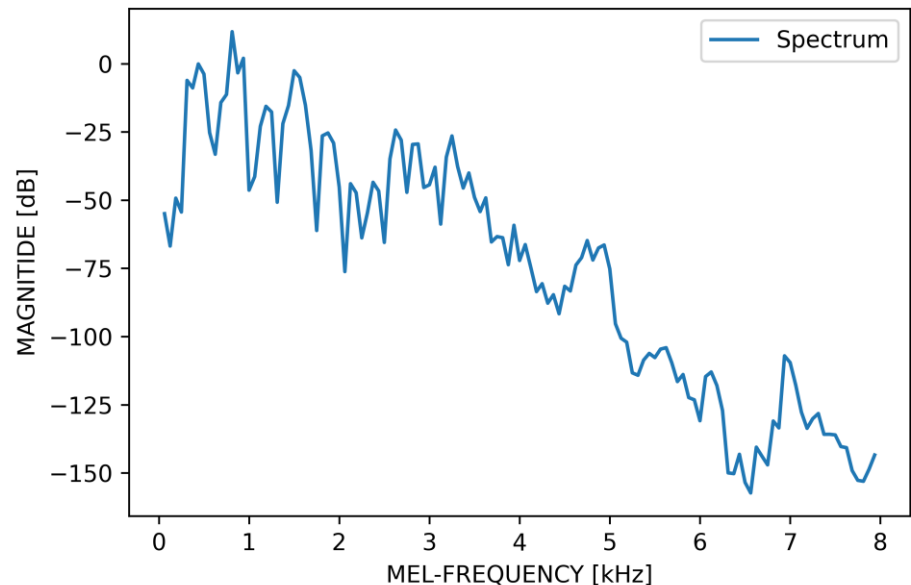
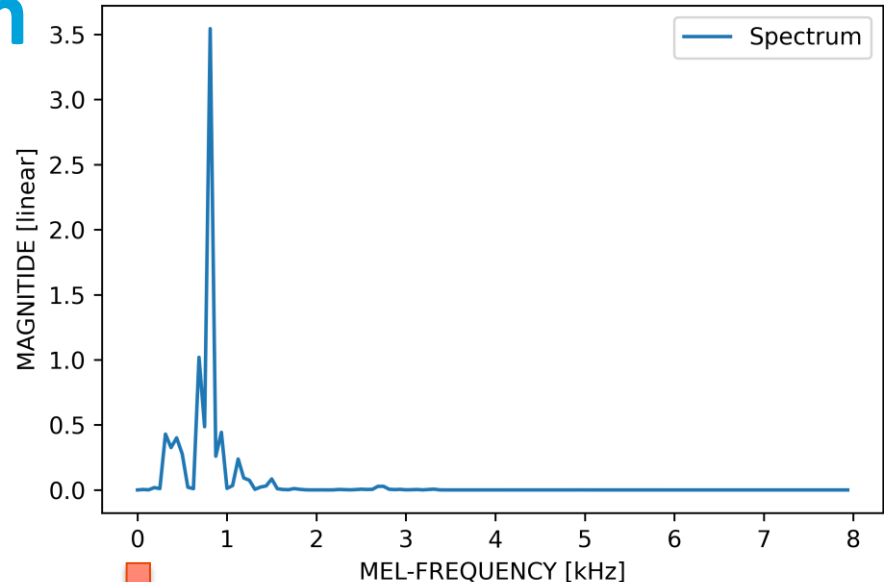
Only 13% of the data

$$IFFT(|X_k| - i|X_k|) \rightarrow \text{Speaker Icon}$$

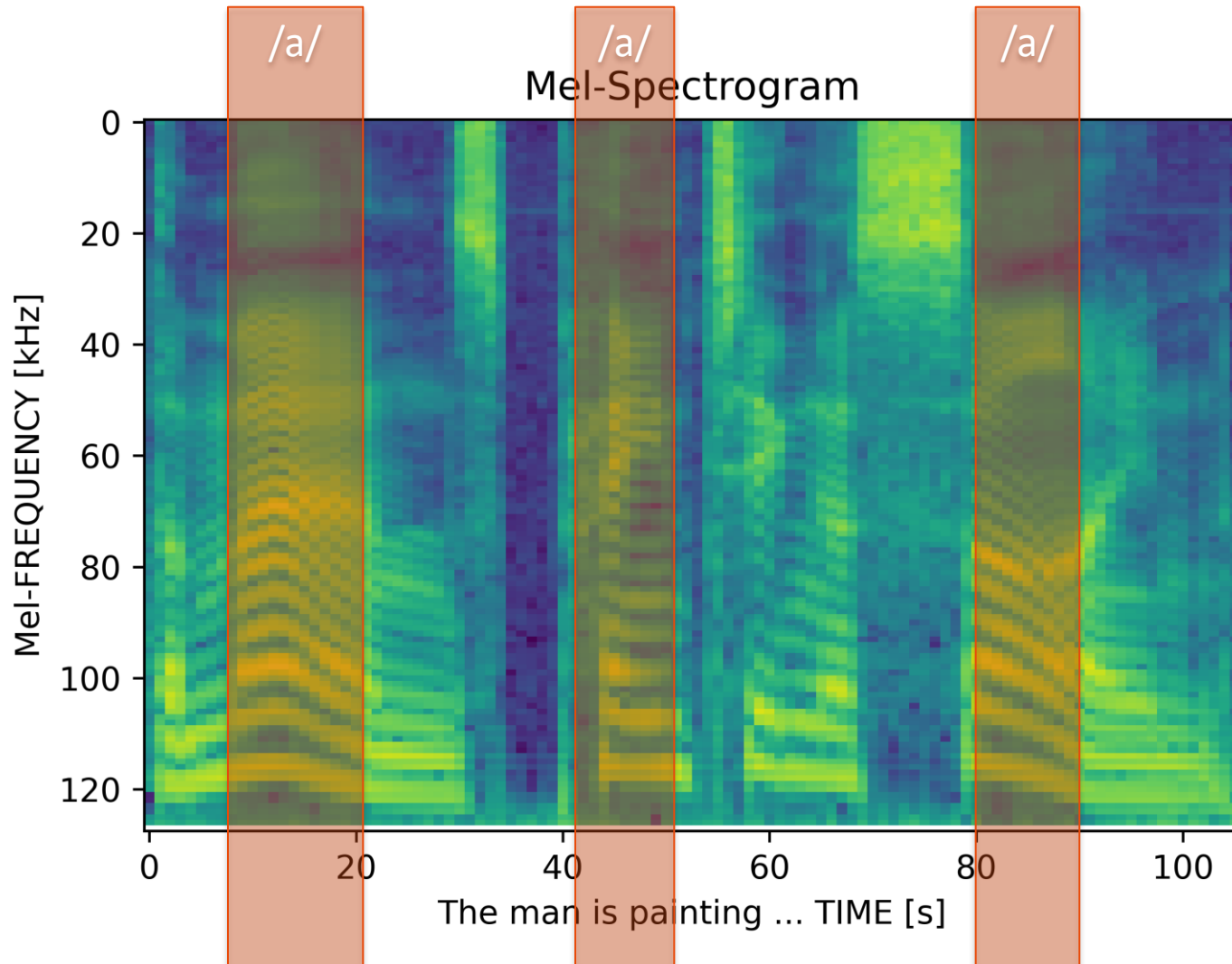


Log Mel Spectrogram

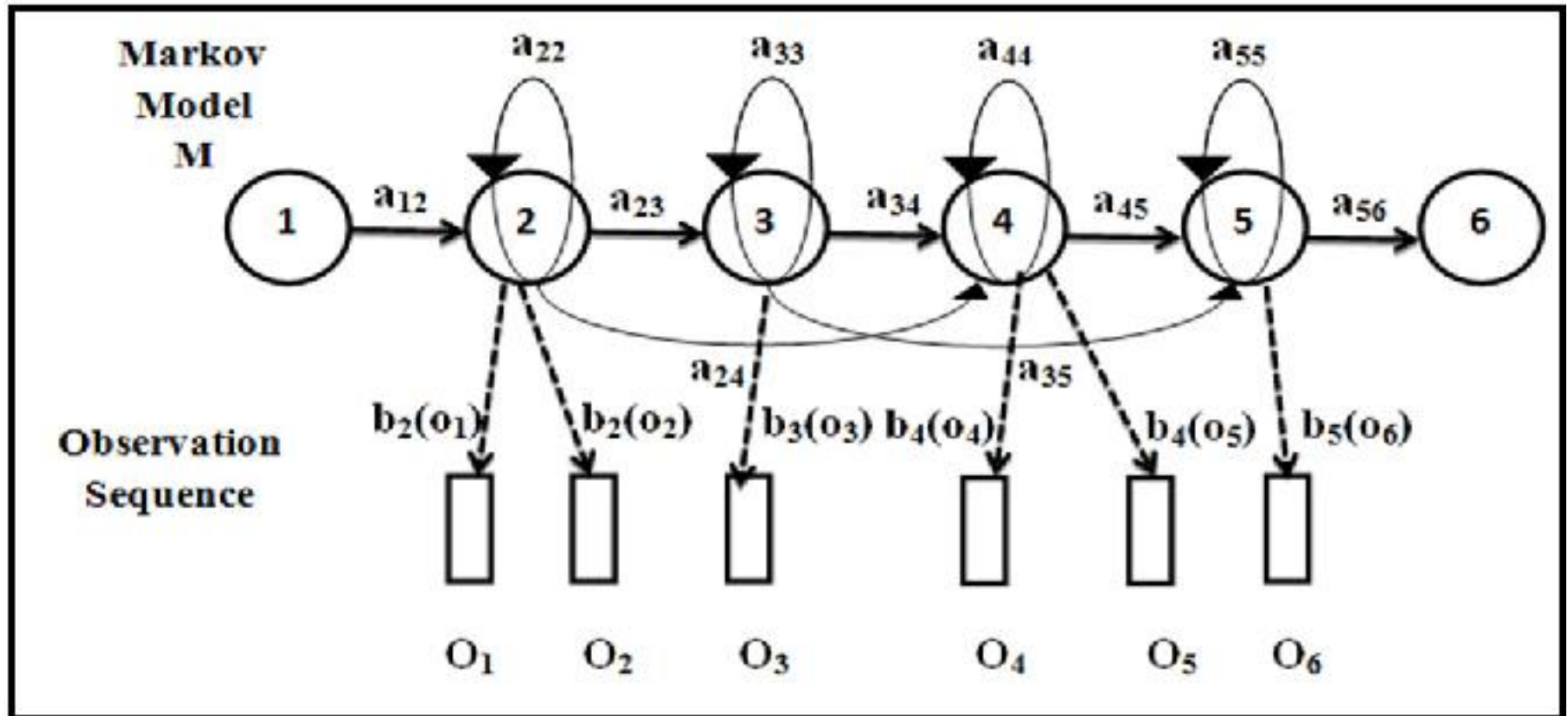
- Our perception of magnitude of a stimulus follows a logarithmic scale (*Fechner's law, 1860*)
- **Spectrum magnitudes are mapped to a logarithmic Decibel scale**
- Aka the **Log Mel Spectrogram**
- Matches with:
 - Frequency resolution and loudness perception of hearing
 - Harmonic spectrum of speech production



Speech vs. text - continuous vs. discrete



Hidden Markov Model (HMM)

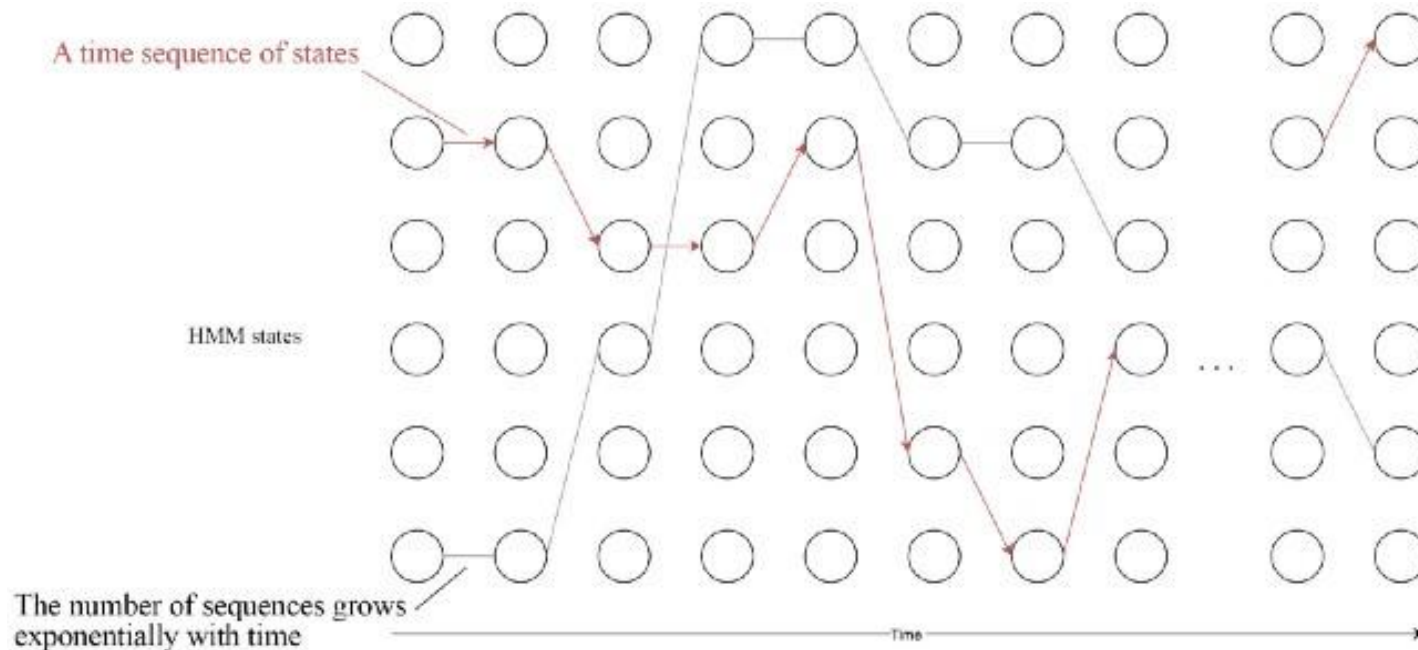


The observations $O_1 \dots$ can be seen as “tokens” – a learned dictionary of LogMelSpec blobs

The “Hidden Model” has less states than #tokens – one hidden stat may emit multiple tokens

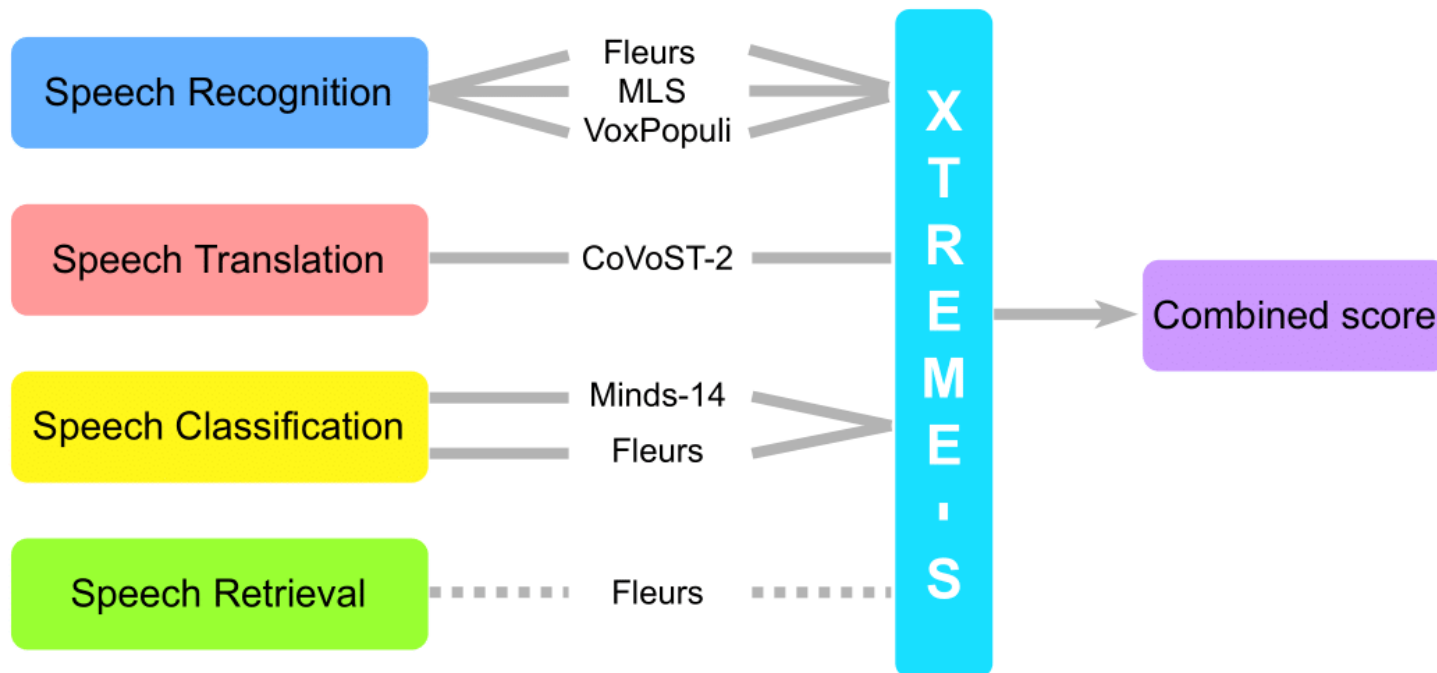
HMM and the Viterbi Algorithm

- Find the most probable path in the alternative state (“token”) sequences, given the observations.



Performance of ASR

- Word Error Rate – %WER = $(S + D + I) / N$
- Character Error Rate - %CER
- BLEU, METEOR, COMET, chrF
- Common benchmarks: FLEUR, ...



Deep learning and ASR



Maastricht University

Speech Recognition & Deep Learning

- Representation
 - Recurrent Neural Networks (RNN) and RNN-Transducer
 - Convolutional Neural Networks (CNN)
 - Self-attention neural networks (Transformers)
 - Variational Autoencoders (VAE)
- Alignment
 - Connectionist Temporal Classification (CTC)
- Other representations
 - i-vectors, x-vectors for speaker recognition, state recognition, health sensing

Classic pipeline

Representation

Acoustic modeling

Language modeling

Search /Alignment

Classic ASR systems (1980's-to-2010's)



Fast Fourier Transform (FFT) - speech to spectrum

Mel-spectrum mapping (and Mel-cepstrum by another FFT) - to auditory representation, Decibel magnitude (logarithm)

TRAINED WITH LARGE COLLECTION OF TRANSCRIBED SPEECH DATA

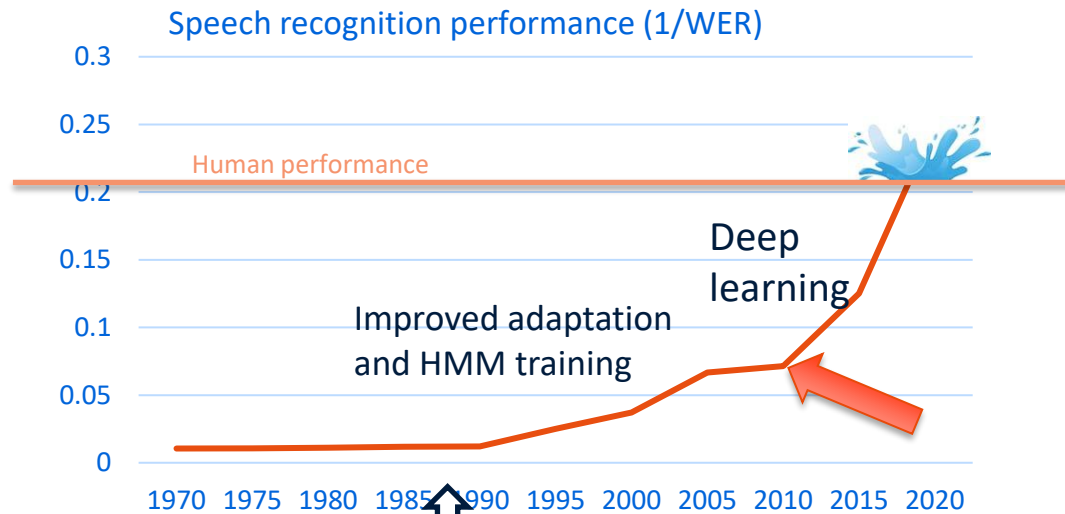
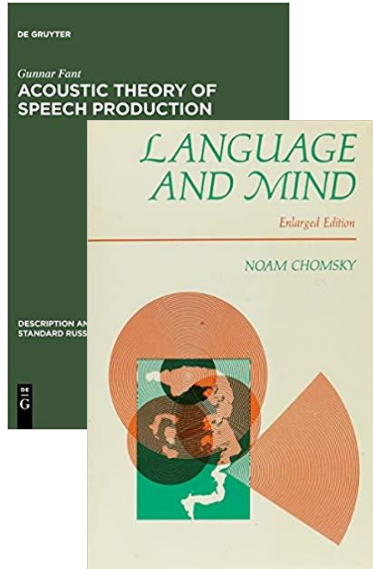
Model spectra by Gaussian Mixture Models (GMMs) – model spectrum “states”

Hidden Markov Model (HMM) – model state transitions by a probabilistic model

Decoding of the most likely phoneme sequence by Viterbi algorithm

“The man is painting a sign”

Deep learning for ASR



Heuristic systems based on language understanding, phonetics

“Every time we fire a linguist, the performance of our system goes up”
(Fred Jelinek, 1988)

Data-driven systems based on learning and a massive search

ASR use cases

- On-edge IoT ASR
 - Speech commands, Q&A
- Personal device ASR
 - Speech input, dictation, Q&A
- Cloud-based GPU-farm premium ASR
 - Critical speech transcription
 - Diarization (meeting minutes)



ASR use cases

OUR TUTORIAL 9

Real-time,
streaming
ASR

Hello thermostat,

When did I last renew my
depression medication?

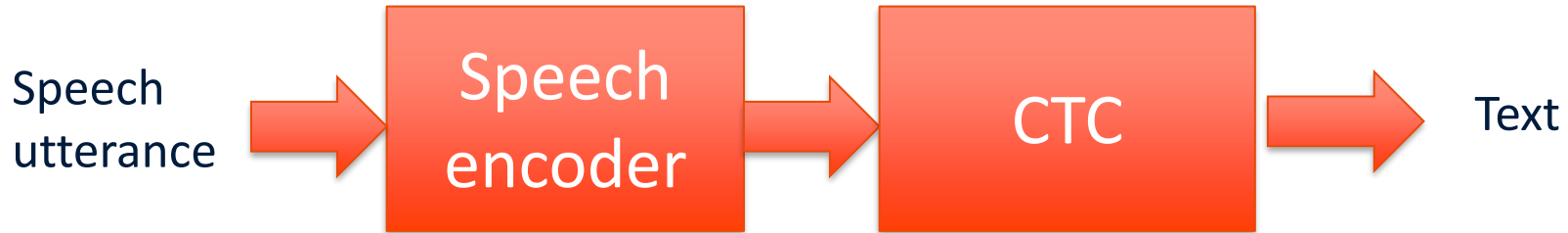
I want to dictate an email to my boss:
“Hi Bob, this is urgent, I propose that ...”

Offline
transcription

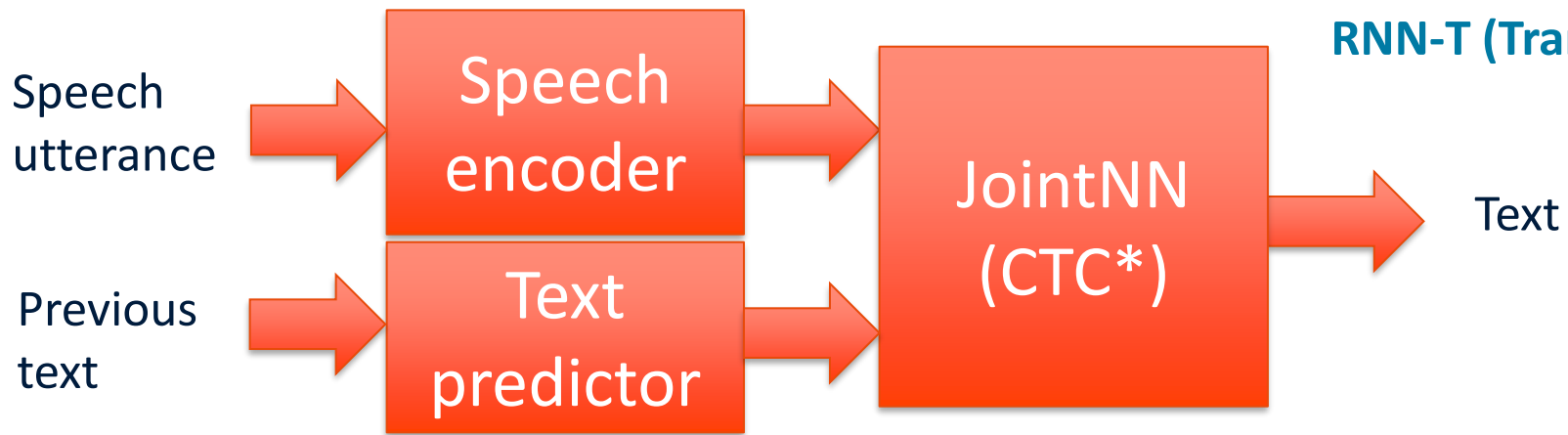


Architectures

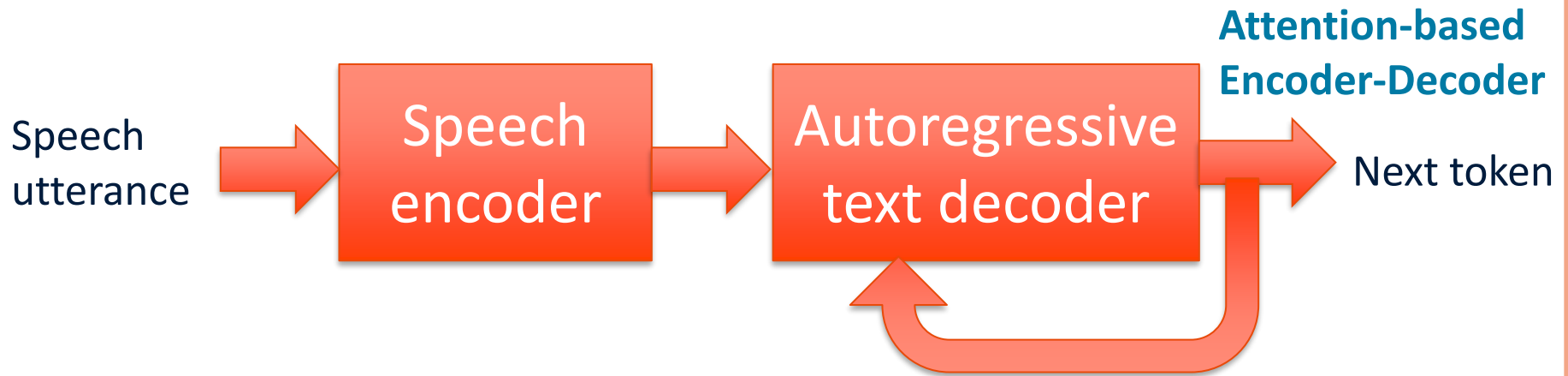
CTC



RNN-T (Transducer)



Architectures



Arch	Use case	Alignment	Performance	Complexity
CTC ASR	On edge/streaming	Explicit (CTC)	Good WER	Low
RNN-T	Streaming	Explicit	Better WER	Medium
AED	Offline (long input)	Implicit	Best WER	High

Popular AED architectures



Maastricht University

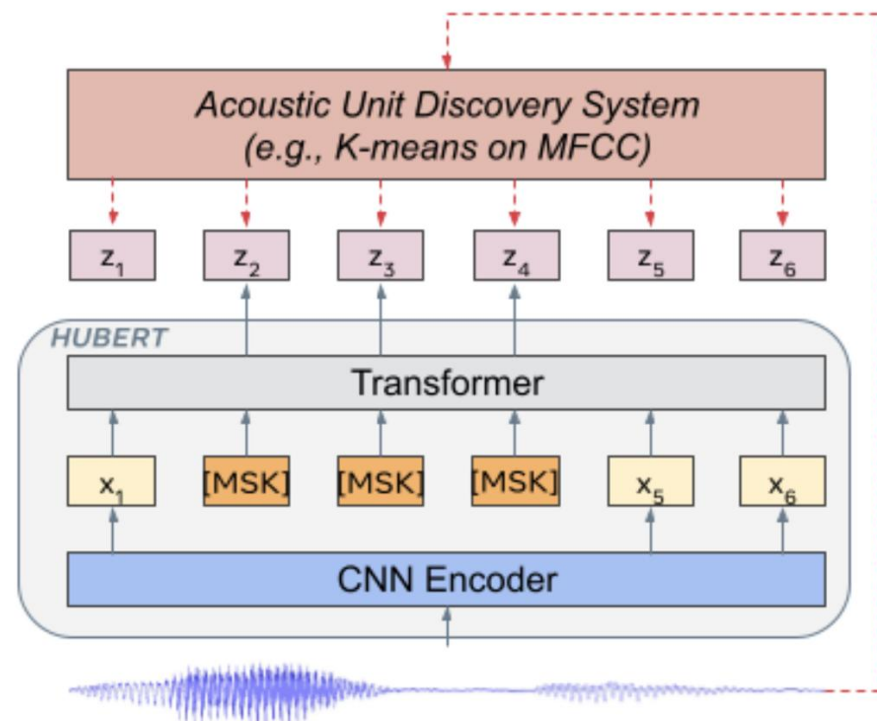
HuBert speech encoder – like BERT for speech

Spectrogram representation of audio

HuBert (a speech model)

- Mel-spectrogram
- CNN encoder to vectors
- Transformer on vectors

Trained in a self-supervised task



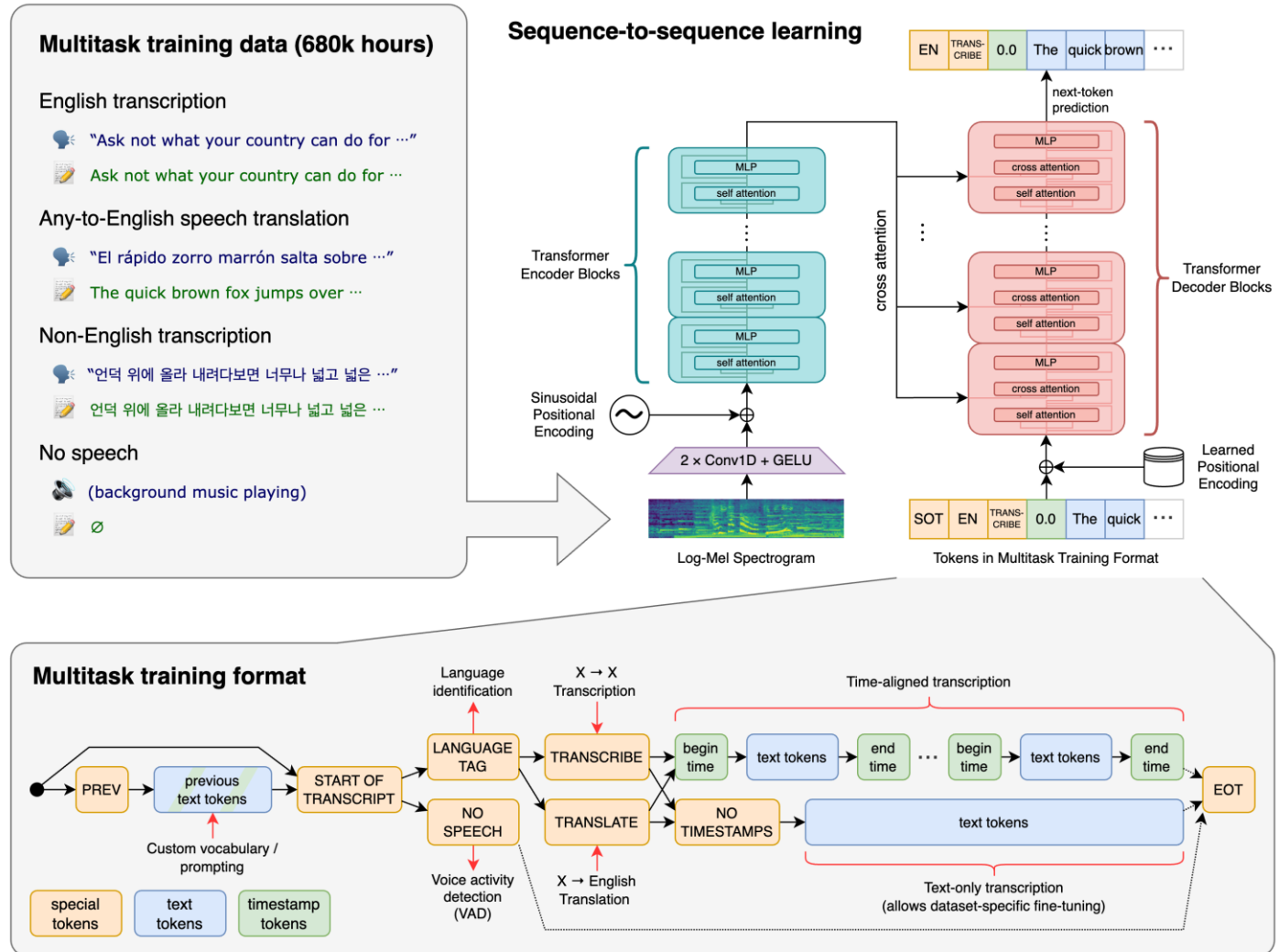
| q_{79} | q_{688} | q_{584} | q_{386} | q_{335} | q_{810} | q_{603} | q_{148} | q_{452} | q_{243} | q_{186} | q_{630} | q_{807} | q_{222} | q_{38} | q_{614} | q_{442} | q_{862} | q_{364}



[\[2106.07447\] HuBERT: Self-Supervised Speech Representation Learning by Masked Prediction of Hidden Units](#)

OpenAI: Whisper

OUR TUTORIAL 9

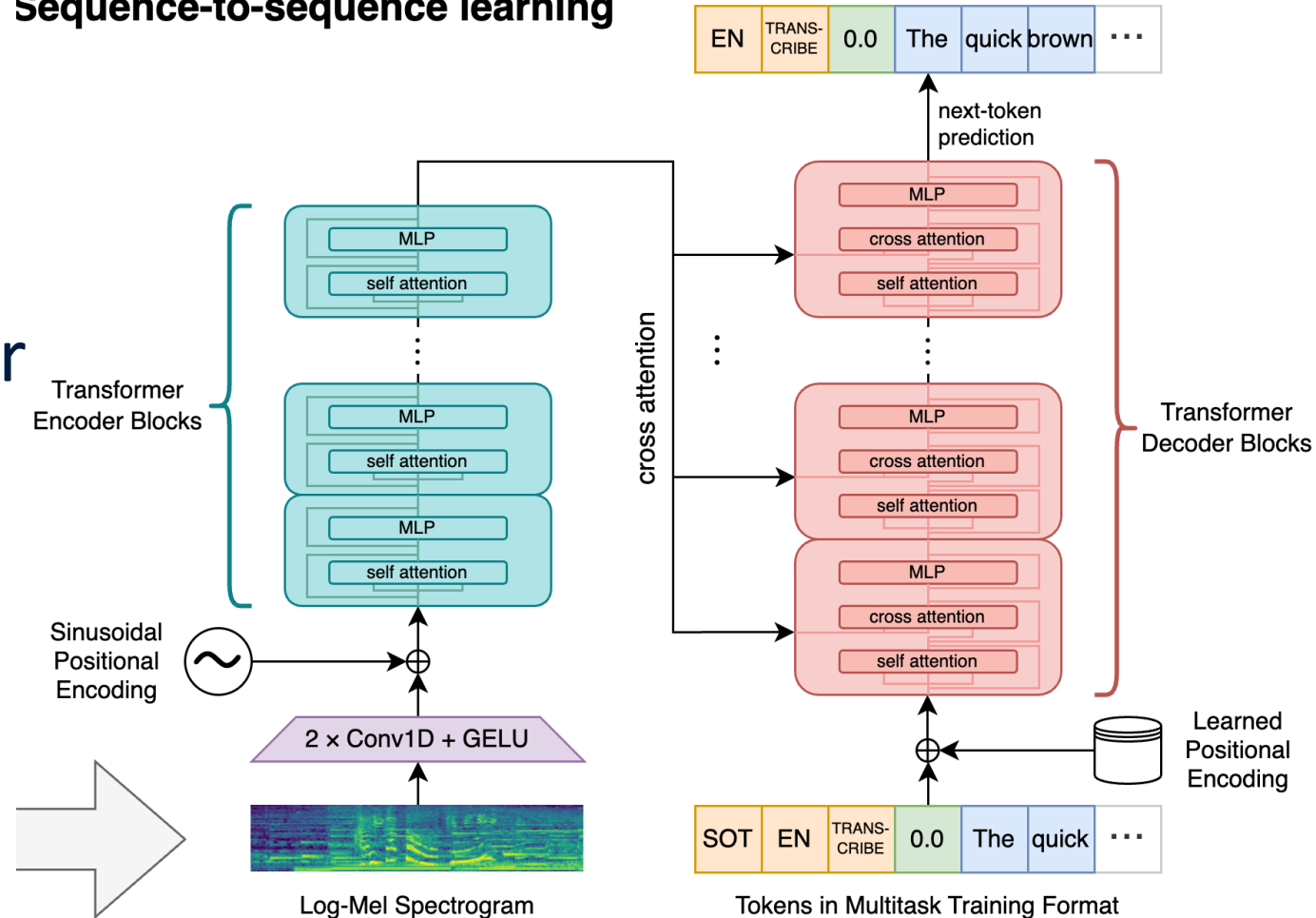


OpenAI: Whisper

OUR TUTORIAL 9

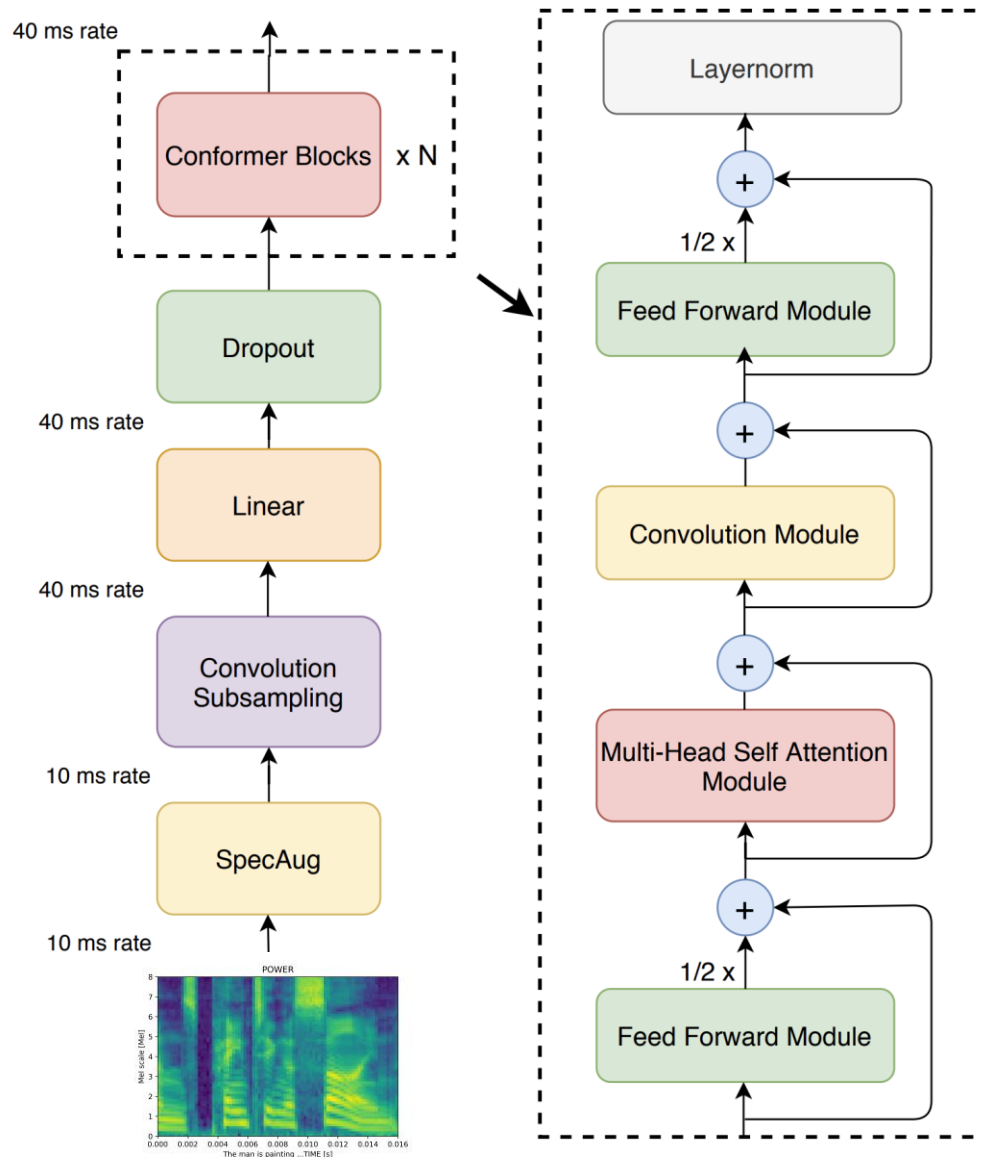
- Encoder-decoder system
- Transformer

Sequence-to-sequence learning

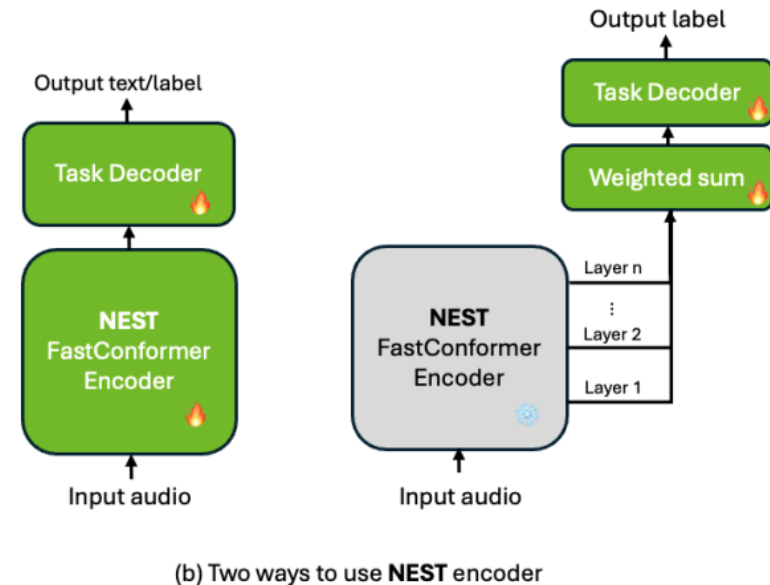
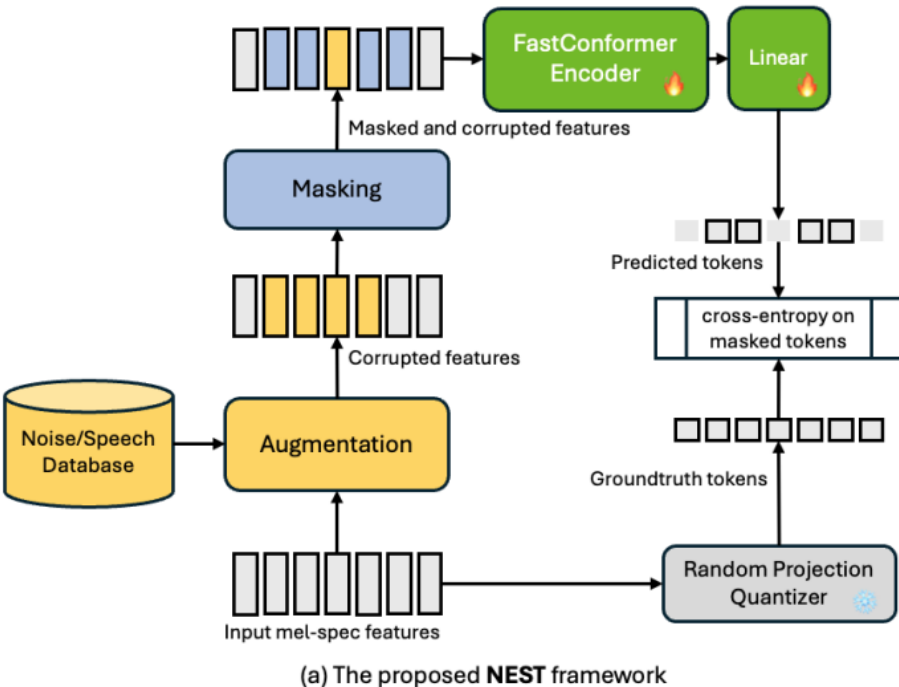


The Conformer (a Transformer-CNN sandwich)

- Transformer is really good in capturing global/long term relations (in text)
- CNN is really good in capturing temporal fine-structures (images)
- Idea: A transformer with CNN modules gets both
- It seems to work better than just a transformer
- Google, AWS, MS use a conformer arch in ASR

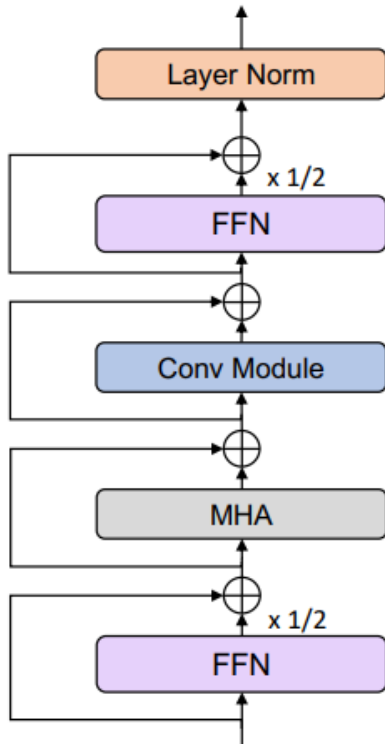


Nvidia Nemo and NEST: Fast Conformer

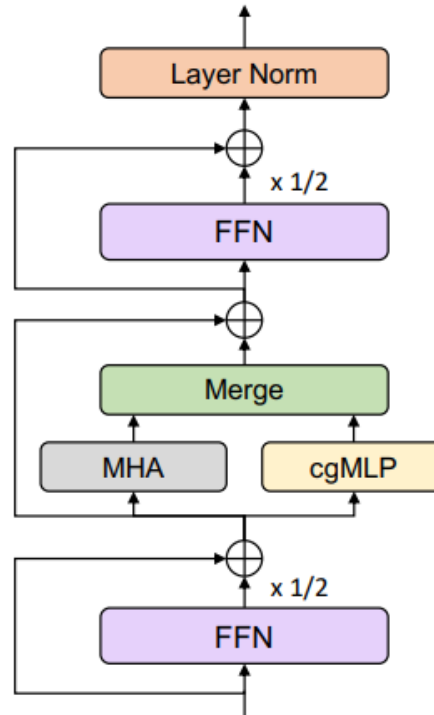


[\[2408.13106\] NEST: Self-supervised Fast Conformer as All-purpose Seasoning to Speech Processing Tasks](#)

E-Branchformer



(a) Conformer [10]



(b) E-Branchformer [13]

Figure 1: Comparison of encoder architectures.

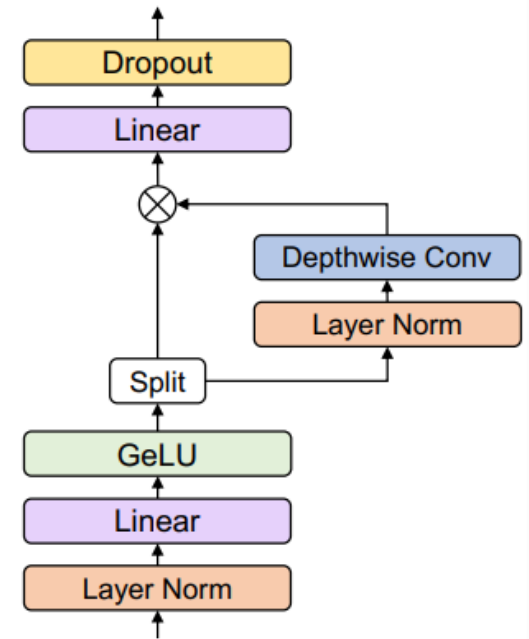
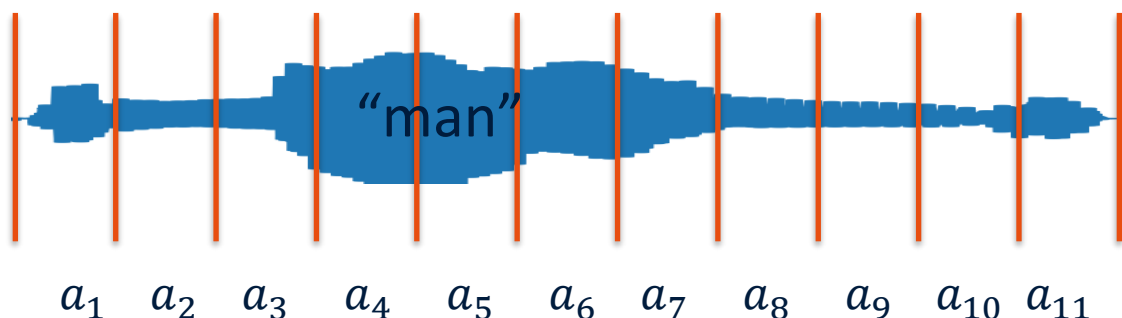


Figure 2: Architecture of cgMLP

From convolution to split conv – MHA

Connectionist Temporal Classification (CTC) – introducing “nothing” $\emptyset$ symbol



m m a a a a a n n n

m $\emptyset$ $\emptyset$ a $\emptyset$ $\emptyset$ $\emptyset$ $\emptyset$ n $\emptyset$ $\emptyset$

Any variant of “man” with 11 symbols
where $\emptyset$ is *blank*

m $\emptyset\emptyset\emptyset\emptyset$ a $\emptyset\emptyset\emptyset$ n
 $\emptyset\emptyset\emptyset\emptyset$ ma $\emptyset\emptyset$ n $\emptyset\emptyset$
 $\emptyset\emptyset$ ma $\emptyset$ n $\emptyset\emptyset\emptyset\emptyset$
 $\emptyset\emptyset\emptyset$ m $\emptyset$ a $\emptyset\emptyset\emptyset$ n $\emptyset$
 $\emptyset$ ma $\emptyset\emptyset\emptyset$ n $\emptyset\emptyset\emptyset$
m $\emptyset\emptyset\emptyset$ a $\emptyset\emptyset$ n $\emptyset\emptyset\emptyset$
m $\emptyset\emptyset$ a $\emptyset\emptyset\emptyset\emptyset$ n $\emptyset$
m $\emptyset\emptyset\emptyset\emptyset\emptyset\emptyset\emptyset$ an
 $\emptyset$ m $\emptyset\emptyset\emptyset$ a $\emptyset\emptyset\emptyset$ n
m $\emptyset$ a $\emptyset\emptyset\emptyset\emptyset\emptyset\emptyset$ n
m $\emptyset$ a $\emptyset\emptyset$ n $\emptyset\emptyset\emptyset\emptyset$
m $\emptyset\emptyset\emptyset\emptyset\emptyset\emptyset$ a $\emptyset\emptyset$ n
m $\emptyset\emptyset\emptyset\emptyset\emptyset\emptyset$ an $\emptyset$
m $\emptyset\emptyset\emptyset\emptyset\emptyset$ a $\emptyset$ n $\emptyset$
 $\emptyset\emptyset\emptyset\emptyset\emptyset$ m $\emptyset\emptyset$ a $\emptyset$ n
 $\emptyset$ m $\emptyset\emptyset\emptyset\emptyset\emptyset$ a $\emptyset$ n $\emptyset$

....

CTC in training



Trans/Conformer



$\mathbf{a} = (a_1 \ a_2 \ a_3 \ a_4 \ a_5 \ a_6)$

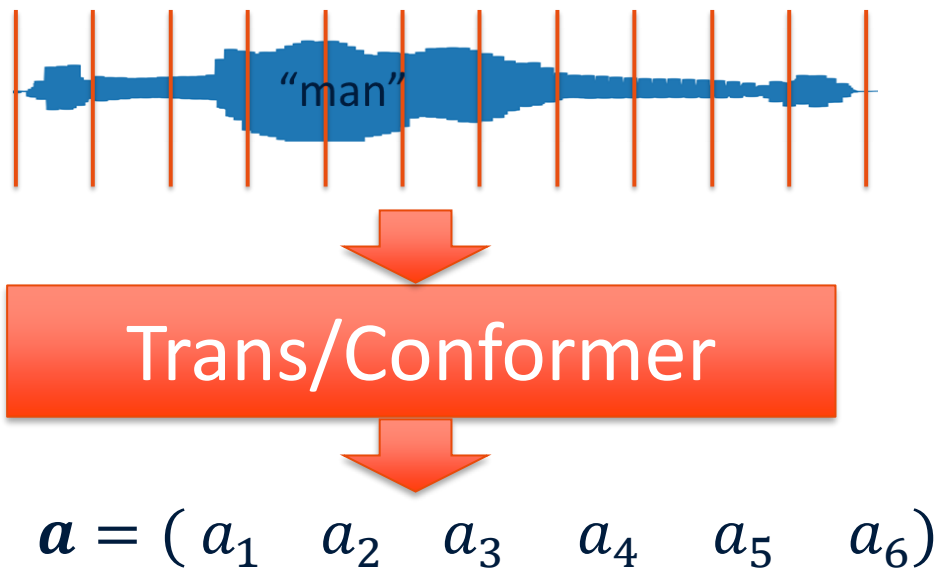
- CTC loss:
$$e = \frac{1}{N} \sum_i^N L(\mathbf{a}, v_i)$$

man

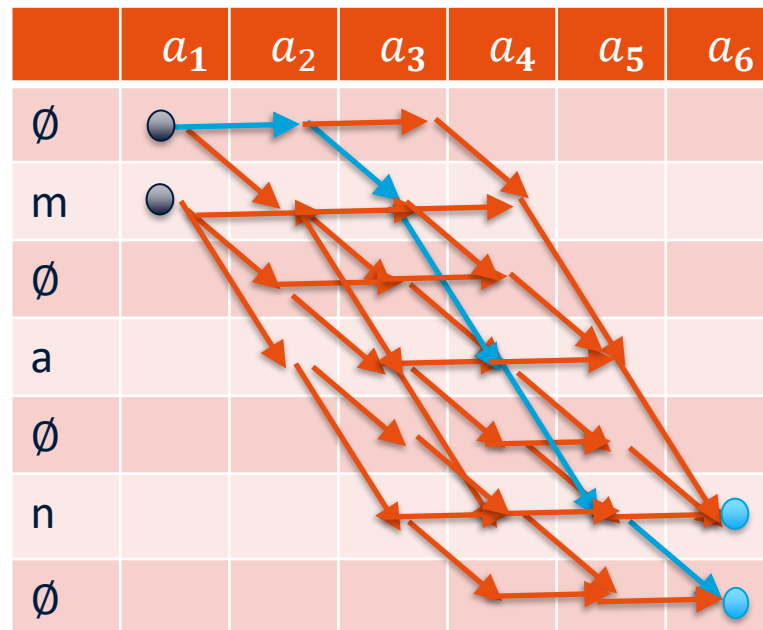
m	a	n	∅	∅	∅	v_0
m	a	∅	n	∅	∅	v_1
m	a	∅	∅	n	∅	v_2
m	a	∅	∅	∅	n	v_4
m	∅	a	n	∅	∅	v_5
m	∅	∅	a	n	∅	v_6
m	∅	∅	∅	a	n	v_7
...						

The loss computation done efficiently by dynamic Programming!

CTC in recognition of one word



Note: each a_i is a softmax vector having probabilities of all tokens
What is the most likely mapping to the text sequence?



$\emptyset\emptyset man\emptyset$

We can compute a Likelihood for each possible alignment and choose the best.
But what about other words?

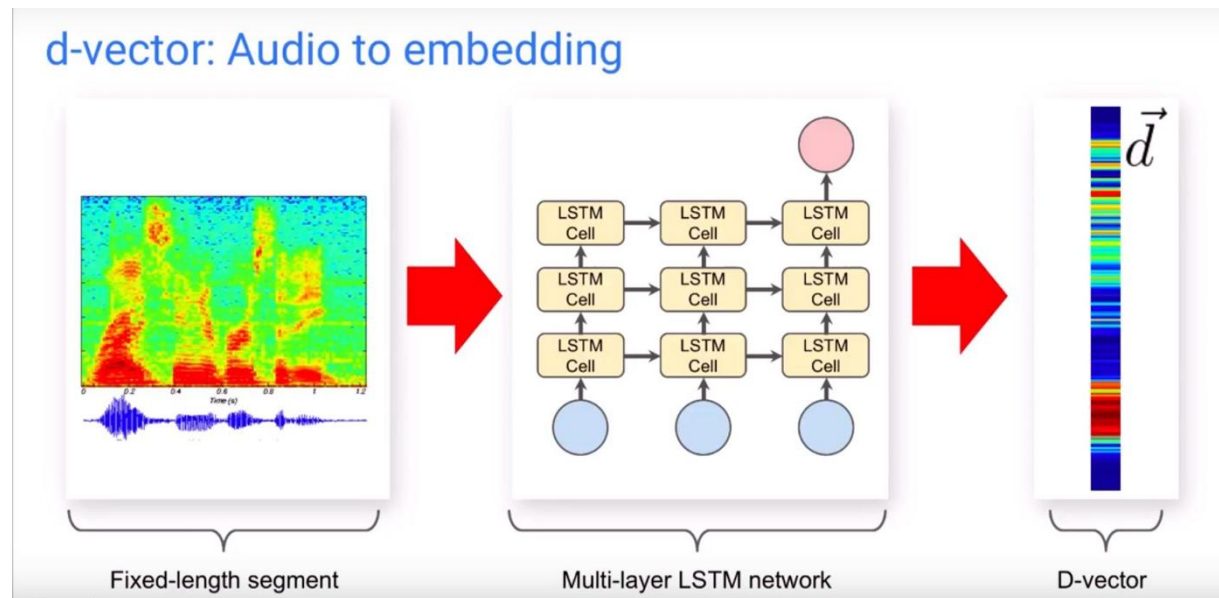
Deep learning beyond ASR



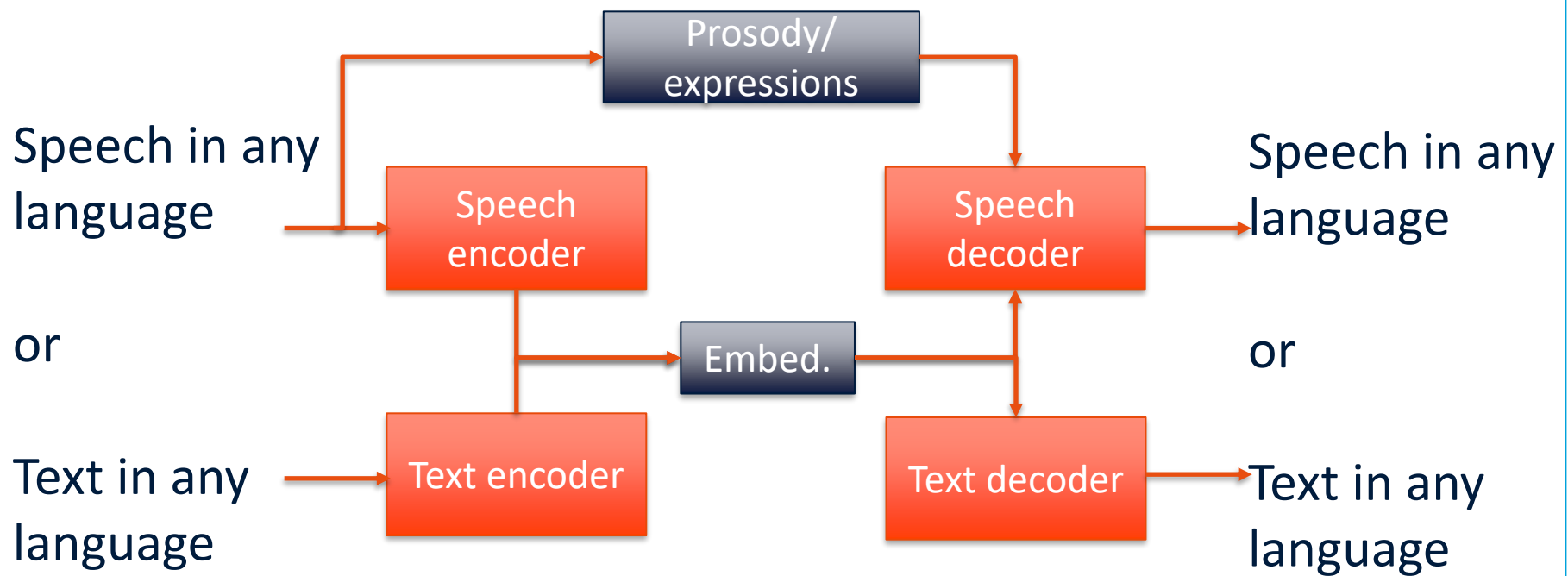
Maastricht University

X-vectors - non-linguistic speech modeling

- In speaker recognition/verification/emotion...
- X-vectors, d-vectors, etc
 - Content-independent neural embeddings for ~3sec speech fragments



Towards universal translation



Meta SeamlessM4T family

Intended Use

- Primary intended uses: SEAMLESSM4T-LARGE v2 *is a multilingual and multimodal translation model primarily for research in speech and text translation. It allows for:*
 - *ASR: Automatic speech recognition for 96 languages.*
 - *S2ST: Speech-to-Speech translation from 100 source speech languages into 35 target speech languages.*
 - *S2TT: Speech-to-text translation from 100 source speech languages into 95 target text languages.*
 - *T2ST: Text-to-Speech translation from 95 source text languages into 35 target speech languages.*
 - *T2TT: Text-to-text translation (MT) from 95 source text languages into 95 target text languages.*
 - *TTS: Text-to-speech synthesis for 36 languages.*

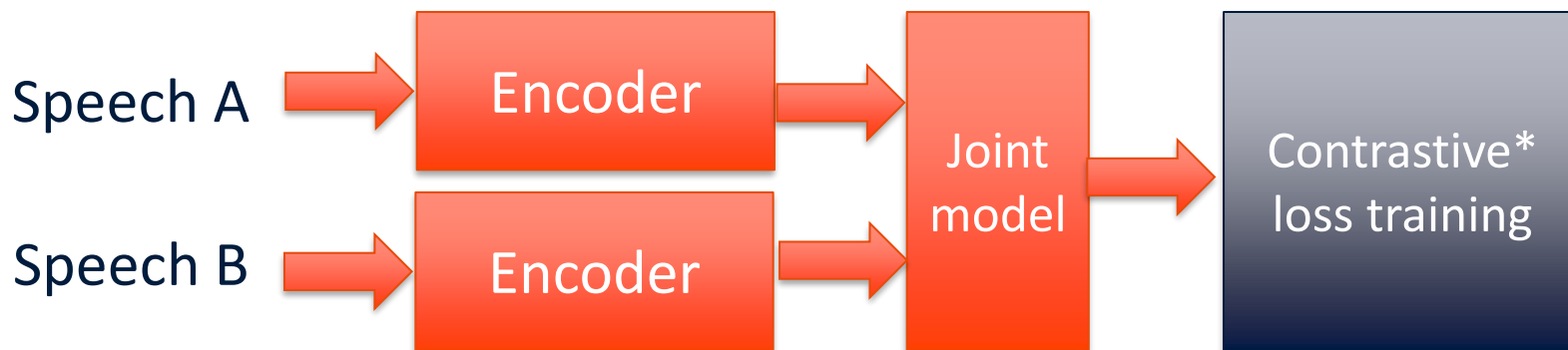


Speech-to-Speech, speech-to-meaning

- Automatic Speech Translation (AST)
 - “no-text” speech-to-speech translation
 - “no-translation” Speech-to-text (other lang.)
 - Speech modification, enhancement
- Speech Understanding
 - Textless understanding of the meaning from speech
- End-to-End speech processing

Speech-to-Speech representations

- “Speech Units”- non-textual embedding of speech sequence for speech seq2seq



Open-source ASRs

Current winners

- NVIDIA Canary
 - IBM-Granite
 - Microsoft Phi-2
 - ...
-
- Whisper

 Spaces |  hf-audio/open_asr_leaderboard   like 1.09k Running on CPU UPGRADE 

Hugging Face is way more fun with friends and colleagues!  [Join an organization](#) [Dismiss this message](#)

Leaderboard



The 🗨️ Open ASR Leaderboard ranks and evaluates speech recognition models on the Hugging Face Hub. We report the Average [WER](#) (⬇️ lower the better) and [RTFx](#) (⬆️ higher the better). Models are ranked based on their Average WER, from lowest to highest. Check the [Metrics](#) tab to understand how the models are evaluated. If you want results for a model that is not listed here, you can submit a request for it to be included  . The leaderboard includes both English ASR evaluation and multilingual benchmarks across the top European languages.

 [Leaderboard](#)  [Multilingual](#)  [Long-form](#)  [Metrics](#) 

model	Average WER ⬇️	RTFx ⬆️
nvidia/canary-qwen-2.5b	5.63	418.28
ibm-granite/granite-speech-3.3-8b	5.74	145.42
ibm-granite/granite-speech-3.3-2b	6	270.57
microsoft/Phi-4-multimodal-instruct	6.02	151.1
nvidia/parakeet-tdt-0.6b-v2	6.05	3386.02
aquavoice/avalon-v1-en	6.24	NA

Open-source ASRs



Real-World Speech-to-text API Leaderboard

This leaderboard compares the best automatic speech recognition (ASR) models and APIs across real world conditions like background noise, non-native accents, and technical vocabulary.

We built this benchmark to find the best speech recognition API for Voice Writer — an AI writing tool that transcribes speech and corrects grammar and wording in real time. Only real-world results matter here, so this benchmark is fully independent and unbiased.

🕒 Leaderboard last updated: October 2025

Category:

Overall

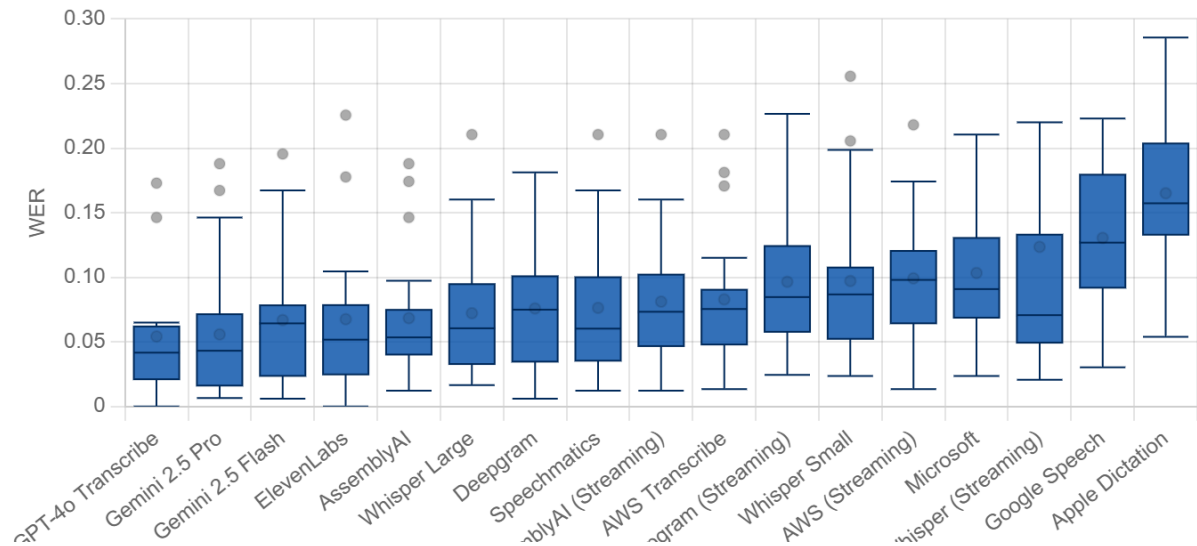


Evaluation: ?

Raw WER



Include streaming ?



Current winners

- OpenAI Transcribe
- Google Gemini
- Eleven Labs
- Assembly AI
- ...
- Whisper



Qwen3-LiveTranslate-Flash

 Response:
你好!

hello

Ciao

こんにちは

Bonjour

Hallo

Здравствуй

สวัสดี

नमस्ते

안녕하세요

Hola

Olá

Understand 18 languages

Multilingual

Vision-Enhanced

Low-Latency

Lossless

 Query:

Ladies and gentlemen, hello everyone. Thank you for your support.
Welcome to Alibaba Group's...(about 1h30min)

 Response:

女士们、先生们，大家好。感谢您的支持。欢迎参加阿里巴巴集团... (大约 1 小时 30 分钟)

女士们、先生们，大家好。感谢您的支持。欢迎参加阿里巴巴集团... (大约 1 小时 30 分钟)

What is mask?

 Query:
This is mask.

 Response:

什么是mask?
这是口罩，
这是面具，
这是面膜。

 Response:

Qwen3-LiveTranslate-Flash真不错!

 Query:
Qwen3-LiveTranslate-Flash is really great.

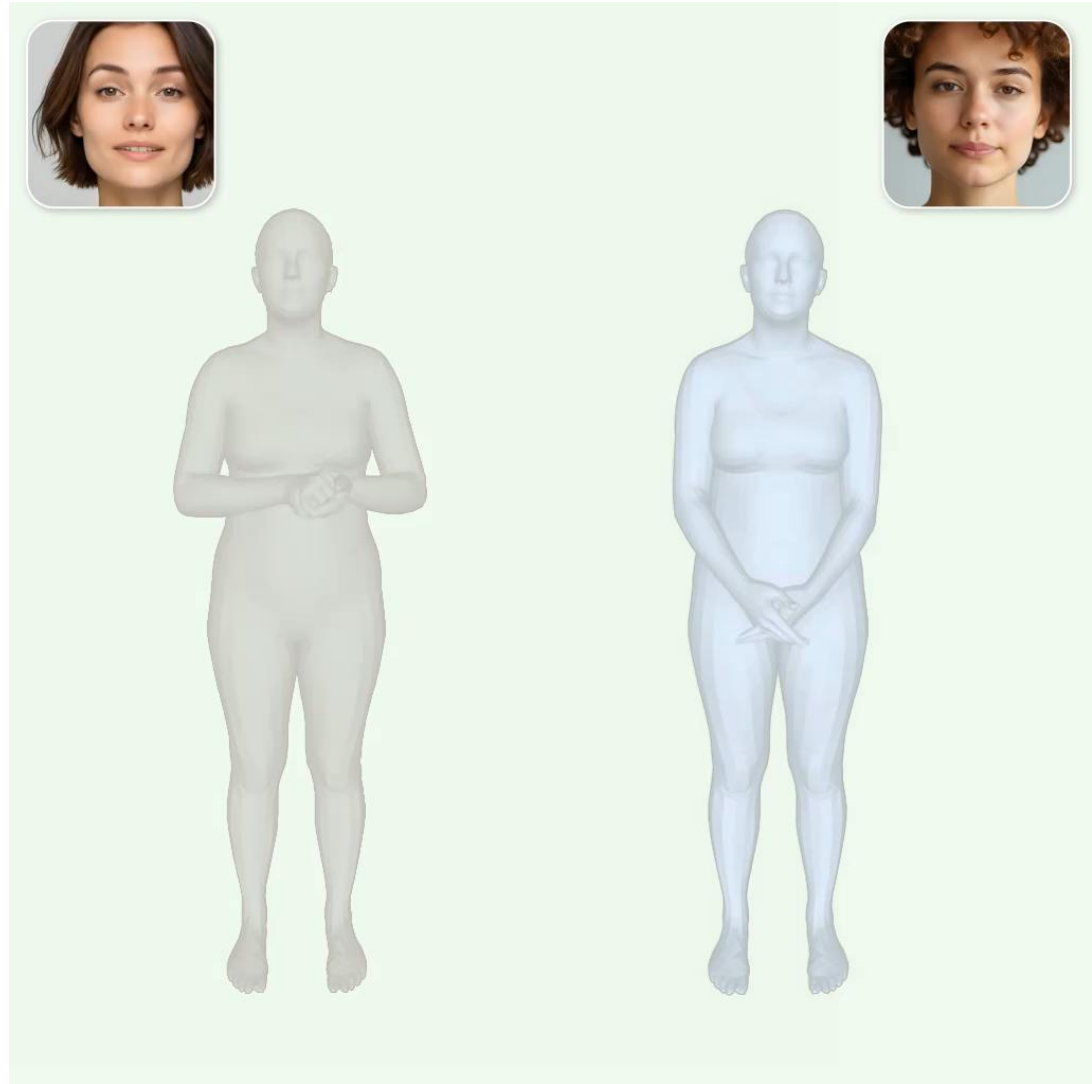
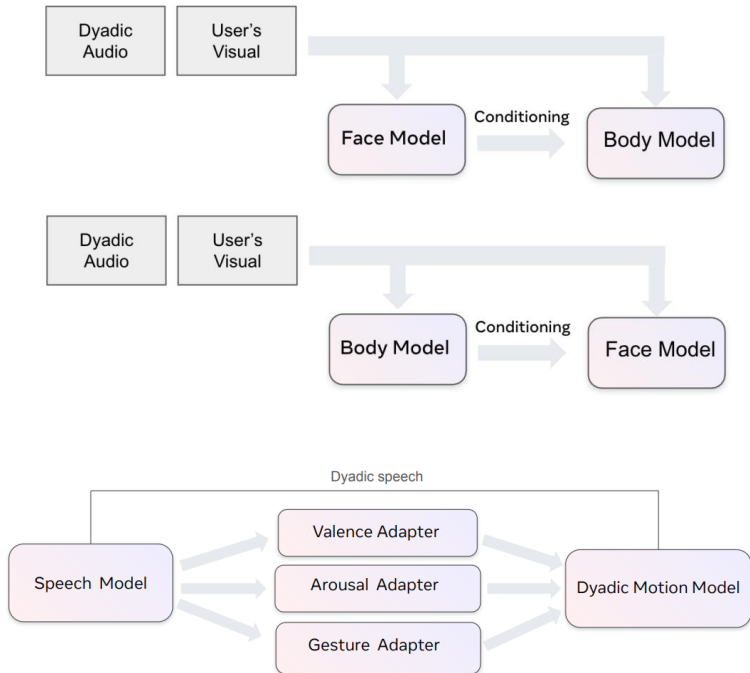
 Response:

Qwen3-LiveTranslate-Flash真不错!

Offline translation

Real-time translation

Speech to body language



Questions?



Maastricht University

Exercise 1: Speech Signal Representation & Log Mel Filterbank Compression

Raw audio is a continuous acoustic waveform that must be windowed, transformed, and mapped to psychophysically motivated representations prior to neural network processing.

- The lecture outlines the windowing process where speech signals are divided into overlapping frames. Explain why speech signals are non-stationary and why frame lengths of 20–25 ms with a 10 ms step size are used instead of analyzing the entire utterance at once.
- In generating a **Log Mel Spectrogram**, explain why the Mel scale applies linear filter spacing below 1 kHz and logarithmic spacing above 1 kHz. Why are spectral magnitudes converted into a logarithmic Decibel (dB) scale based on Fechner's law?
- Using the data compression calculations (before NN):
 - Given high-quality audio at 48 kHz, down-sampled to wideband 16 kHz (16,000 samples/sec).
 - If you compute a Mel-spectrum using 64 filter bands with a 10 ms hop step (yielding 100 frames/sec), verify the resulting data rate in numbers/second and explain why this preserves only ~13% of the original data.

Exercise 2: End-to-End ASR Architectures (CTC, RNN-T, and Attention-based Encoder-Decoder)

This lecture introduces three modern end-to-end Automatic Speech Recognition (ASR) paradigms: CTC, RNN-Transducer, and Attention-based Encoder-Decoder (AED).

- Complete a comparison across the three architectures addressing:
 - **Alignment Strategy:** Which rely on explicit frame-level alignment versus implicit cross-attention alignment?
 - **Latency & Streaming:** Which are suited for on-edge streaming applications, and which are typically restricted to offline batch transcription of fixed audio chunks?
 - **Word Error Rate (WER) vs. Complexity:** How do they rank in relative complexity and WER?
- We also introduce **Connectionist Temporal Classification (CTC)**. Explain the functional role of the "blank" symbol $\emptyset$. Using the word "*man*", show how multiple valid 11-symbol alignment sequences (e.g., repeating tokens and blanks) map to the same final transcription.
- What architectural design makes **The Conformer** (a "Transformer-CNN sandwich") superior to a pure Transformer for acoustic modeling? What does the CNN component capture compared to the multi-head self-attention module?

Exercise 3: Classic ASR vs. End-to-End Speech Models & Paralinguistic Representation

Modern speech architectures have evolved from modular statistical systems (GMM-HMM) into unified, multimodal foundation models (Whisper, Meta SeamlessM4T).

- Review the classic ASR pipeline from the 1980s–2010s:
 - What were the distinct roles of the **Acoustic Model** (GMMs/HMMs) and the **Language Model** decoded via Viterbi?
 - How does an end-to-end model like OpenAI's **Whisper** replace this modular pipeline using a single Transformer encoder-decoder?
- What is the standard evaluation metric for ASR accuracy (Word Error Rate - WER)? How is WER computed, and why is transcribing speech to pure text described as a "lossy channel" for human communication?
- Beyond literal text transcription, explain the role of **x-vectors** and self-supervised **Speech Units**. How do these representations enable tasks like speaker verification, emotion detection, and direct "no-text" speech-to-speech translation without generating intermediate text transcripts?

Discussion



On the Text-First Bias: Did NLP Build on the Real Language, or on Its Residue?

Speech is 1.5 million years old; writing is 5,000. Almost everything in this course, from grammars to GPT, was built on the younger one. Writing keeps the words and drops the rest: pitch, pace, hesitation, emphasis, who is speaking and how they feel.

Is text a representation of language, or an artefact that language left behind, which we then mistook for the thing itself?

If most of what a listener receives never reaches the transcript, what does a text-only model understand when it understands a conversation? And what would change in NLP if speech, not text, were the default?

On Jelinek's Maxim: Can Data Replace the Ear?

“Every time we fire a linguist, the performance of our system goes up” (Jelinek, 1988). For decades speech front-ends were engineered from the biology of hearing: short windows, Mel filters, decibels. Whisper and HuBERT learn from tens of thousands of hours of audio with almost no hand design.

And yet both still start from a log-Mel spectrogram. Did scale make the ear irrelevant, or is the ear the one piece of knowledge that nobody has managed to fire?

What does it mean that our best models of speech begin with a model of the listener? Is that an inductive bias we could drop with more data, or a reminder that speech is defined by being heard?

On Textless Speech: Are Words Necessary for Meaning?

Speech-to-speech translation used to pass through text: recognise, translate, synthesise. Models like SeamlessM4T now go from waveform to waveform through learned speech units, with no words in between, and carry the tone and the voice across.

When you listen, do you turn sound into words before you understand it, or do words come afterwards, when you try to say what you understood?

If a system can translate meaning and feeling without ever producing a word, are discrete symbols necessary for meaning, or a technology for storing and sharing it? What would be lost in a world that no longer needed the transcript?

